

Vectors and Linear Transformations

Ling 282/482: Deep Learning for Computational Linguistics

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Fall 2026

Linear Algebra Objects

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- **Scalars**
 - Single numbers
 - What you're used to elsewhere in math
 - examples: 0, 1, 3.14, π , 7/22

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Vectors

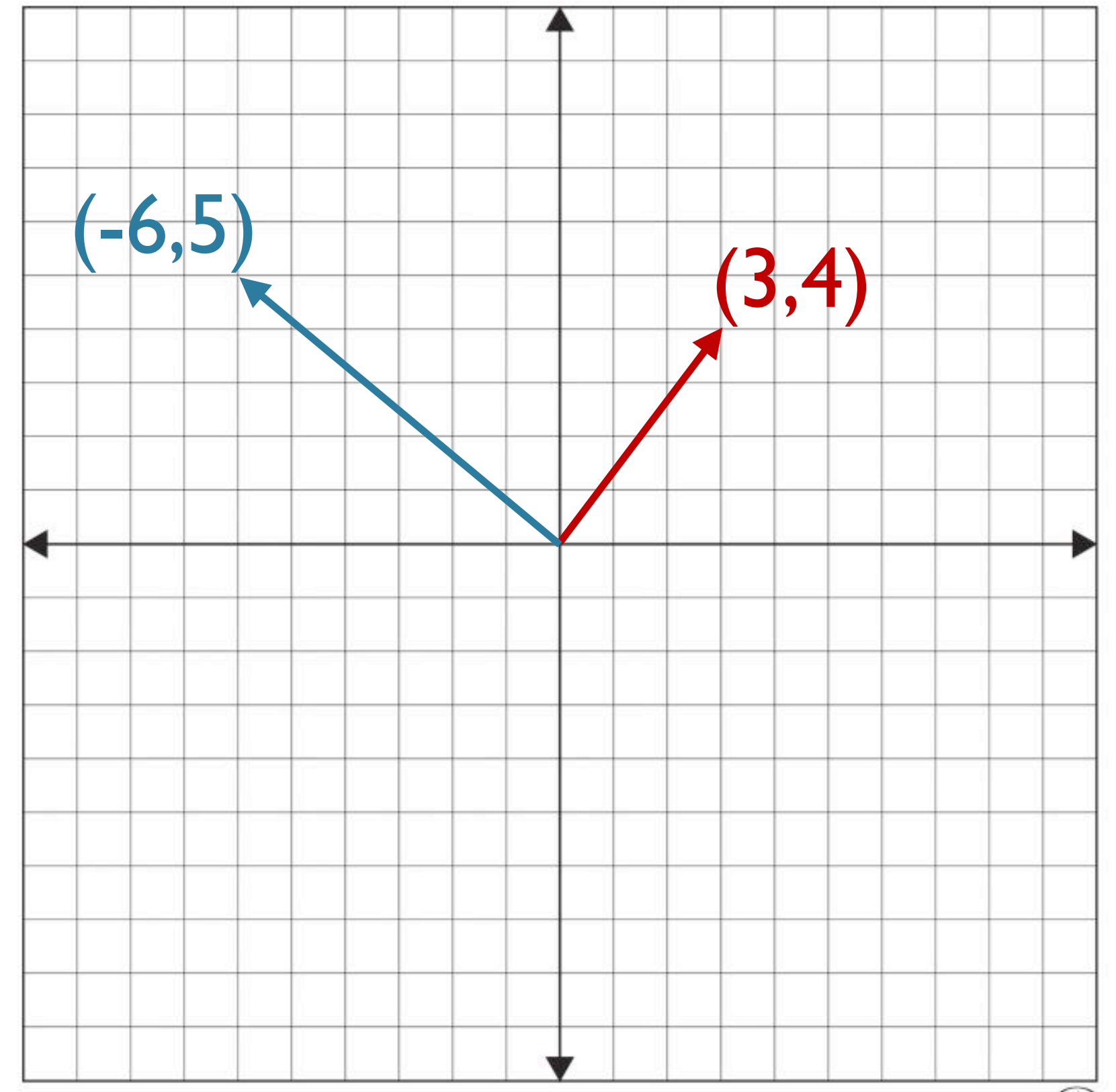
- By default, vectors are considered to be ***columns***
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 - Often visualized as **arrows** or **points** in space

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$$c \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} cx_1 \\ cx_2 \\ cx_3 \end{bmatrix}$$

(c is a scalar)

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- The dot product returns a **single value!!** (not a vector)
- Intuition: the **strength** with which the two vectors **go in the same direction**
 - (Not super important to remember this right now)
 - Important fact: **perpendicular vectors** have a dot product of **zero!**

Vector Spans and Spaces

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- Note: $a = \mathbf{0}$ is used to indicate a vector of zeros

Vector (In)dependence Examples

- What constants solve this equation?

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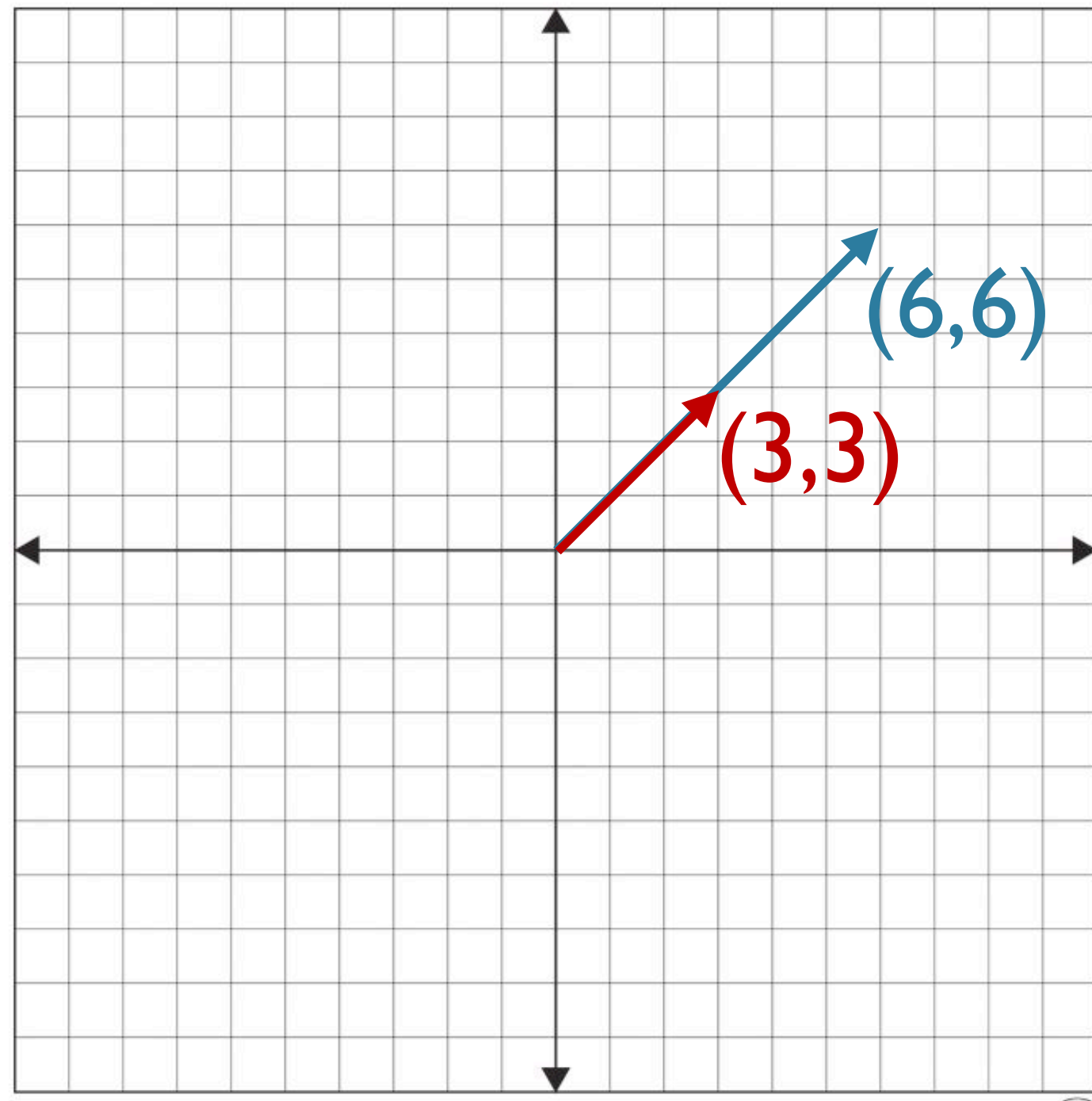
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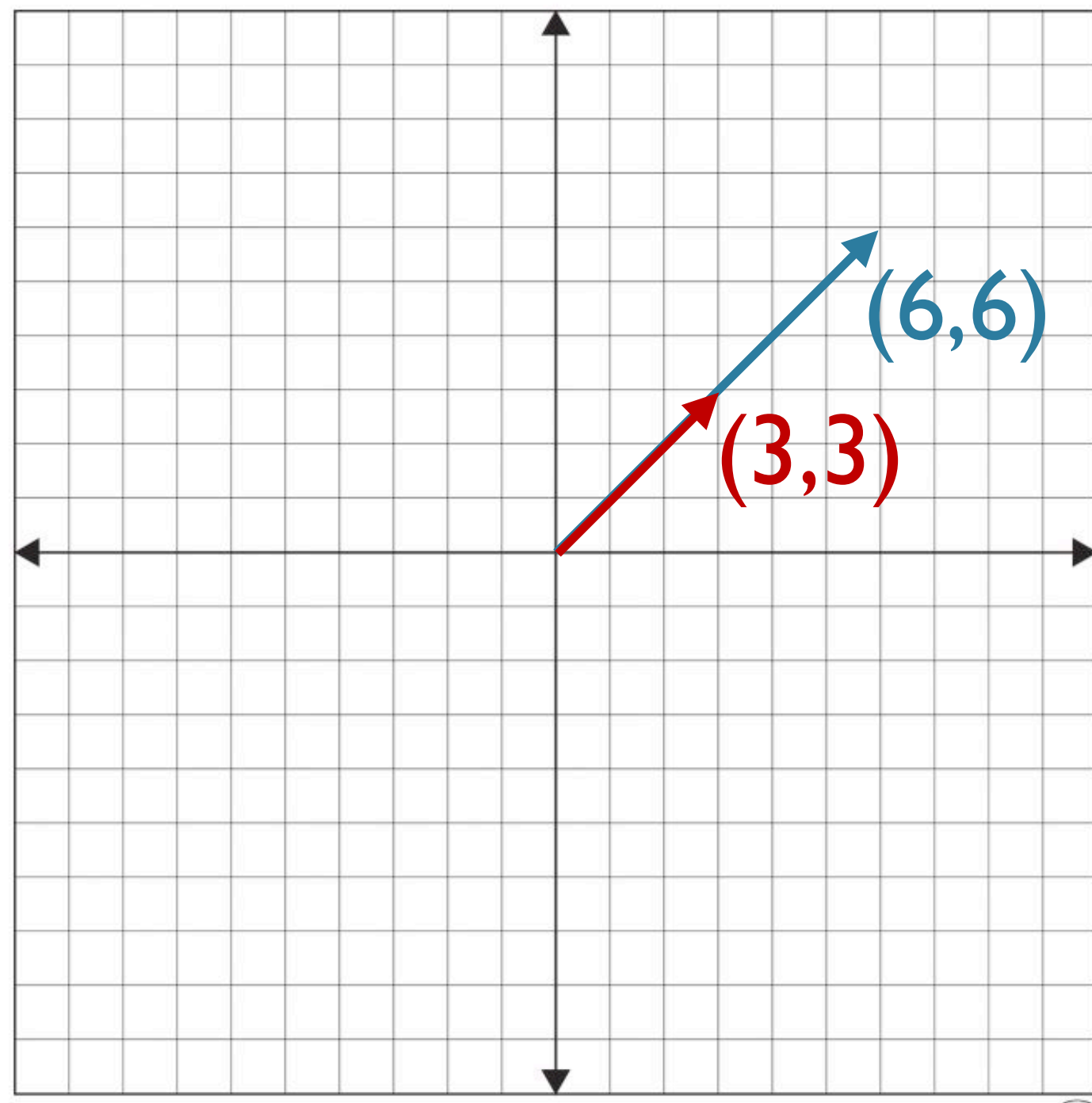
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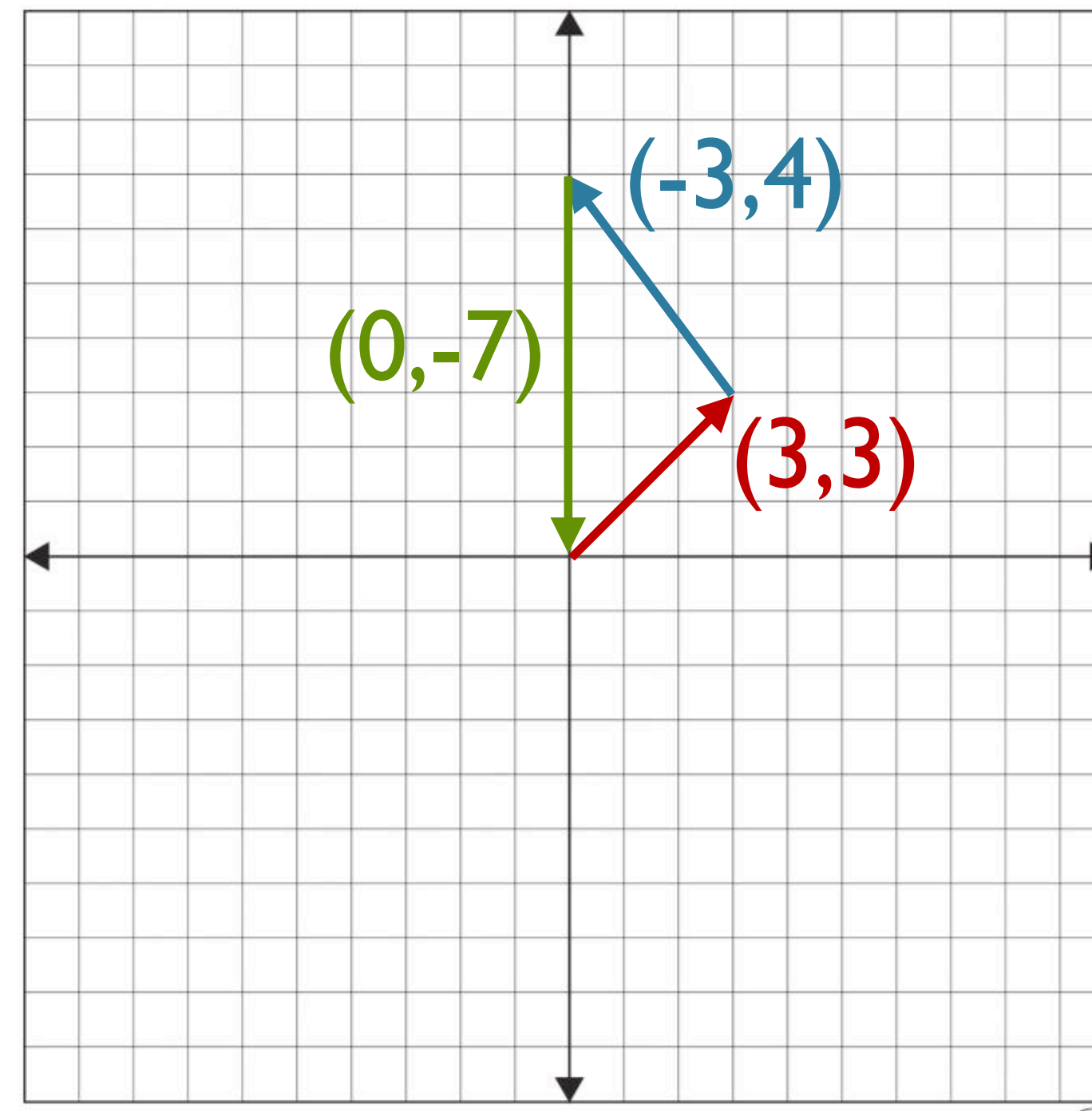
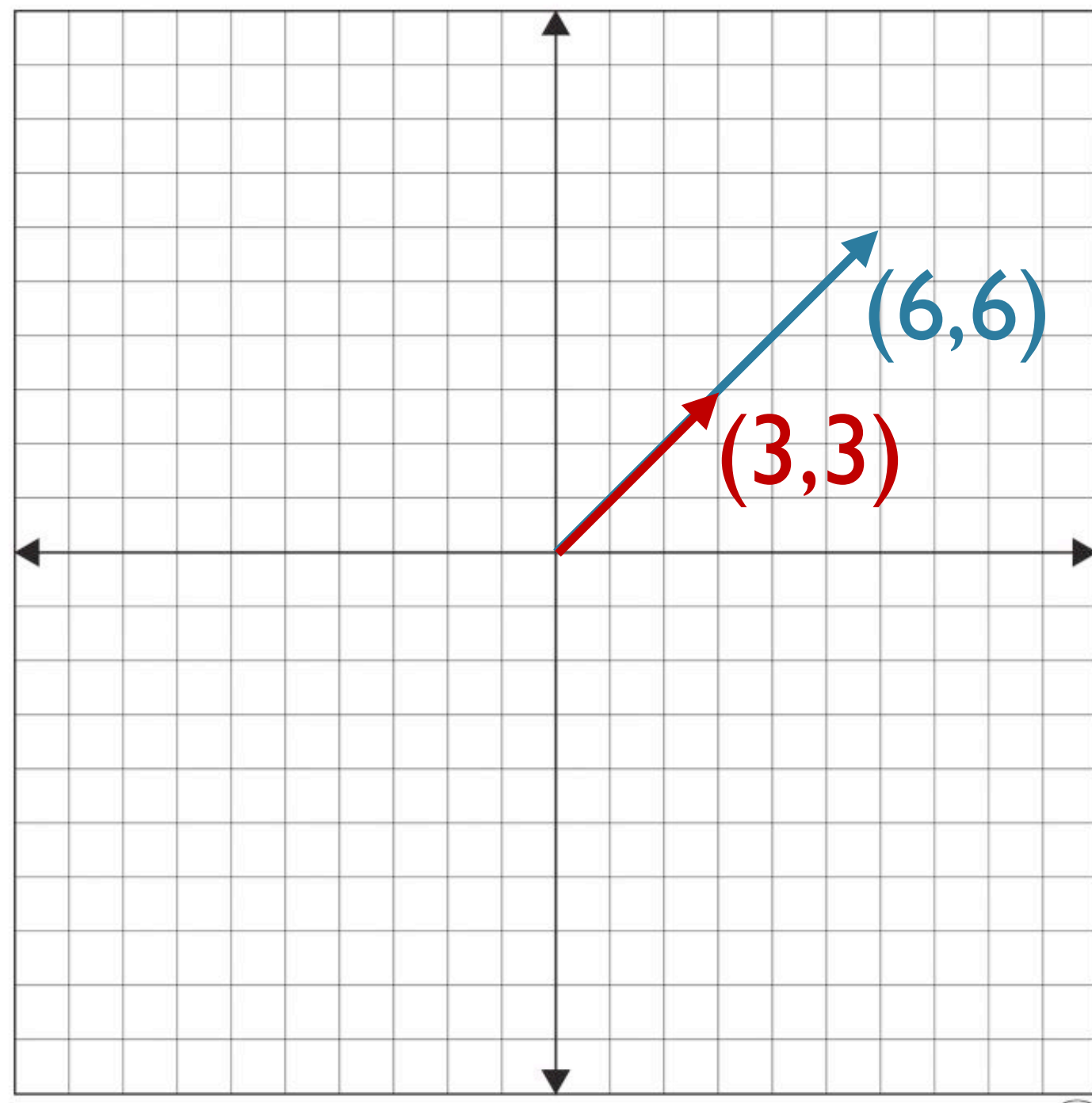
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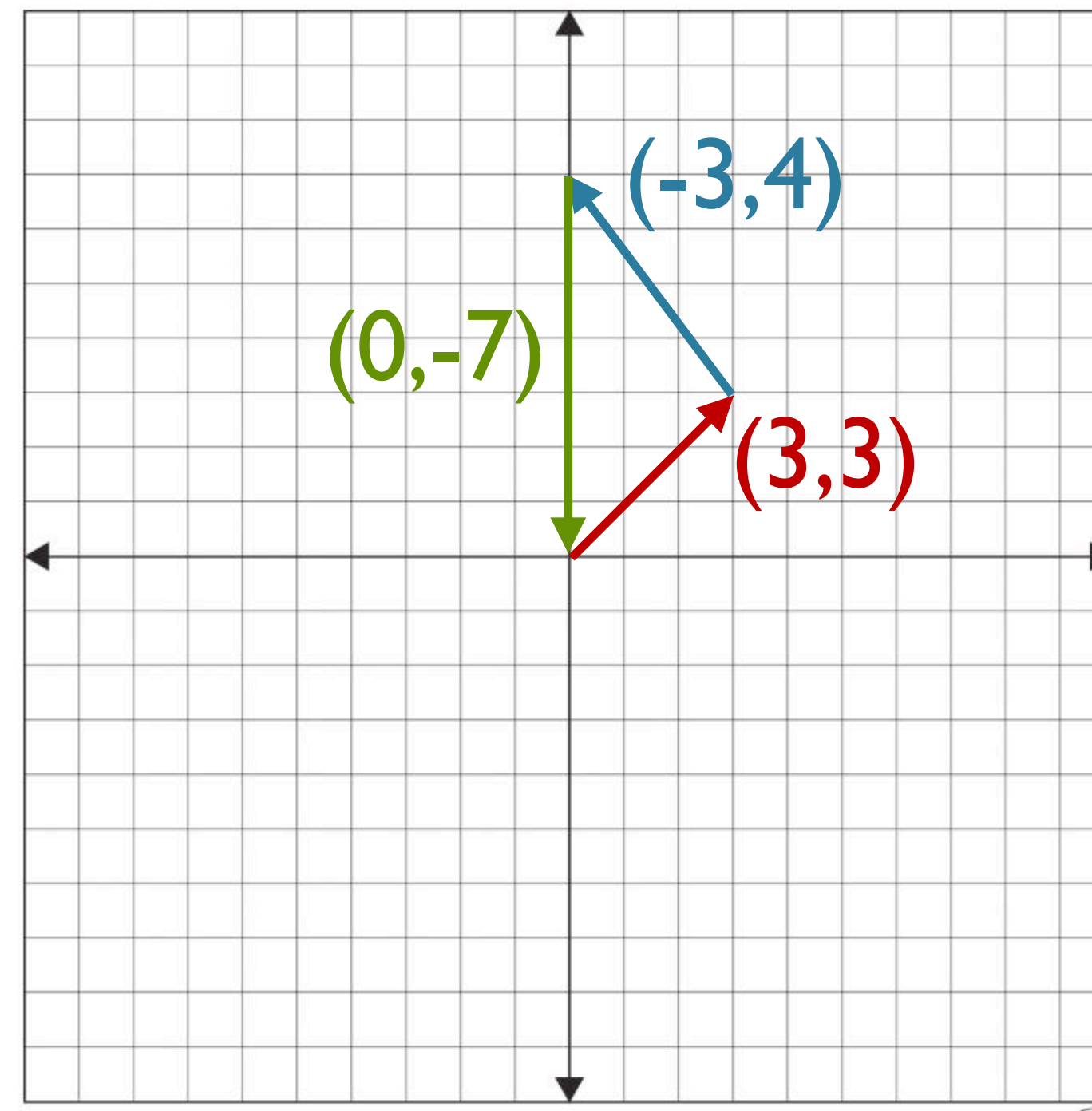
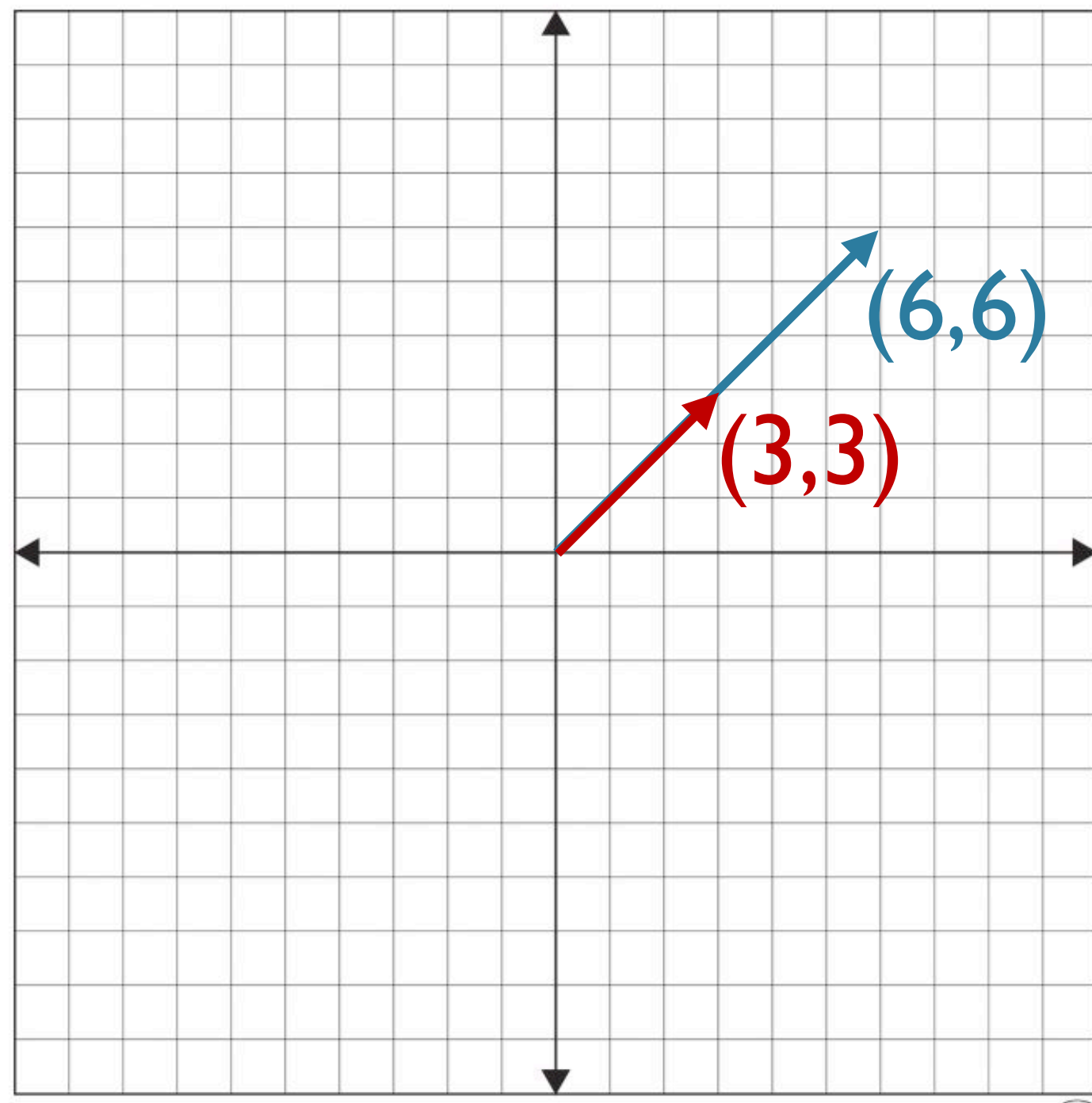
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(this is what
adding
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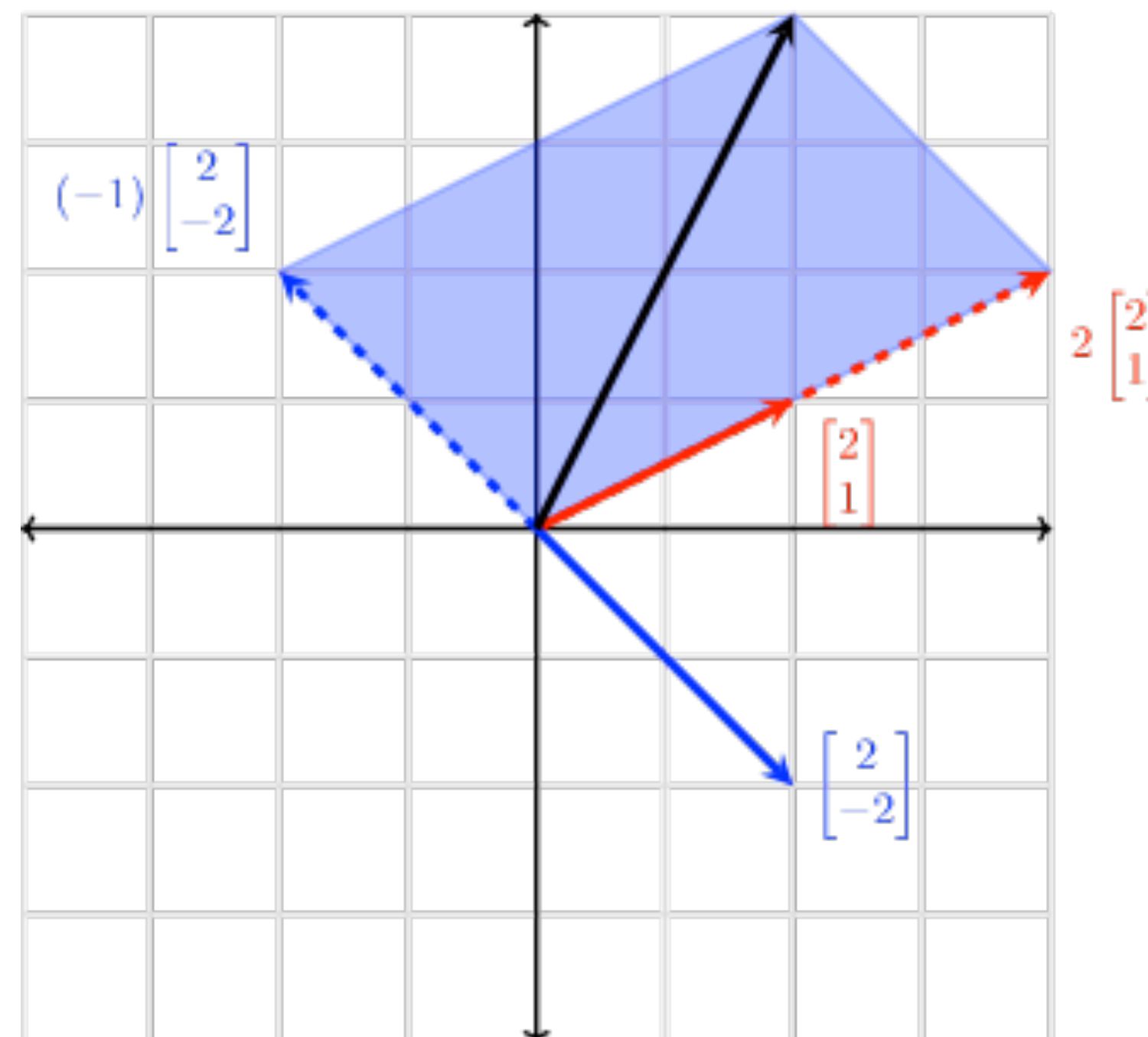
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 - Ex: **a** and **b** above span a **2-D plane** in R^3

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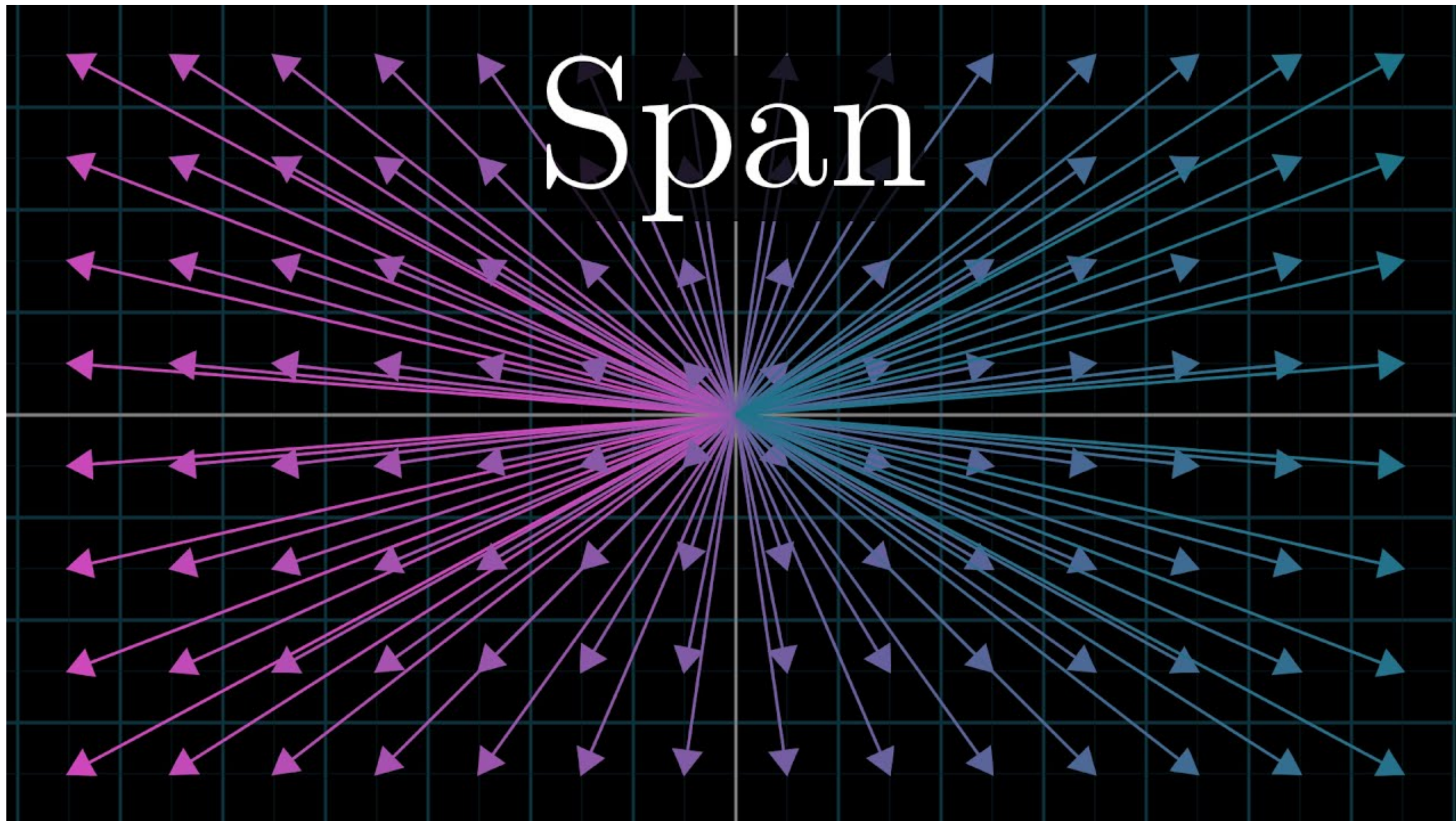
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- These are not the only bases for these spaces

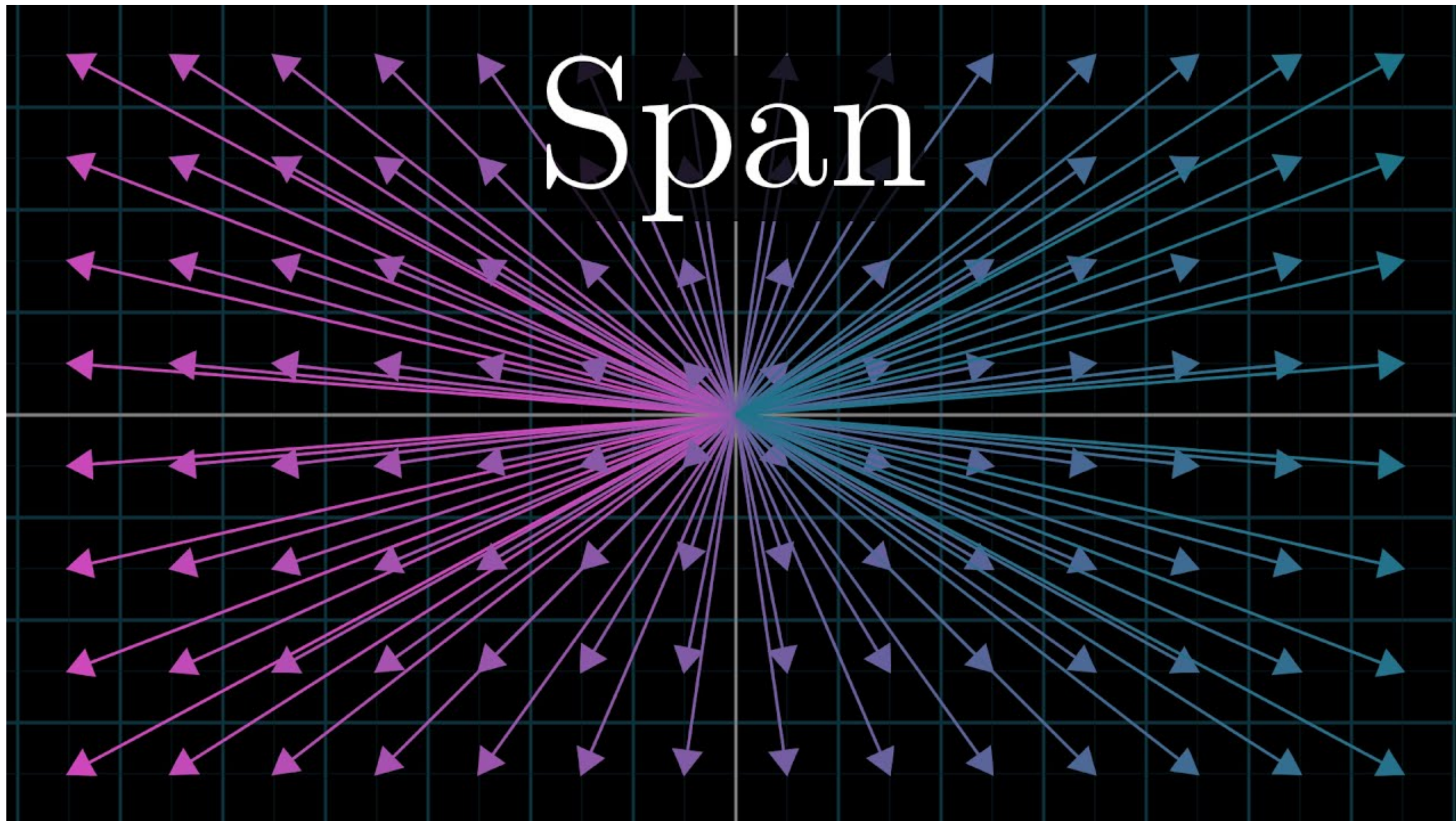
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Span Video



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Matrix Multiplication

Quick reminder: Dot Product

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(vectors need to be the same length)

Matrix-Vector Multiplication

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \quad x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$Ax = ?$$

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$$\begin{array}{c} 4 \text{ rows} \\ \begin{bmatrix} 1 & 5 \\ 2 & 6 \\ 3 & 7 \\ 4 & 8 \end{bmatrix} \\ 2 \text{ columns} \\ \text{“4x2 matrix”} \end{array} \quad \begin{array}{c} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \\ 3 \times 2 \end{array} \quad \begin{array}{c} \begin{bmatrix} 7 & 9 & 11 \\ 8 & 10 & 12 \end{bmatrix} \\ 2 \times 3 \end{array}$$

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Matrix-Vector Multiplication

The Traditional Way

$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} - \\ - \\ - \end{bmatrix}$$

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 - The vector provides the **constants** for a **linear combination of the columns**

$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} \textcolor{red}{1} \\ \textcolor{blue}{1} \\ \textcolor{green}{1} \end{bmatrix} \longrightarrow \textcolor{red}{1} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \textcolor{blue}{1} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + \textcolor{green}{1} \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 12 \\ 15 \\ 18 \end{bmatrix}$$

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 - ... b is always in the **span of A 's columns**
 - This is called the **Column Space of A** , $C(A)$

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- What can you tell about the Column Space of this matrix?

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third column not independent of first two

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- MxN matrix can be considered a **function** from R^N to R^M
 - **However**, the function's range **may not span** R^M , unless it is **rank M**

Linear Transformations

Identity Matrix

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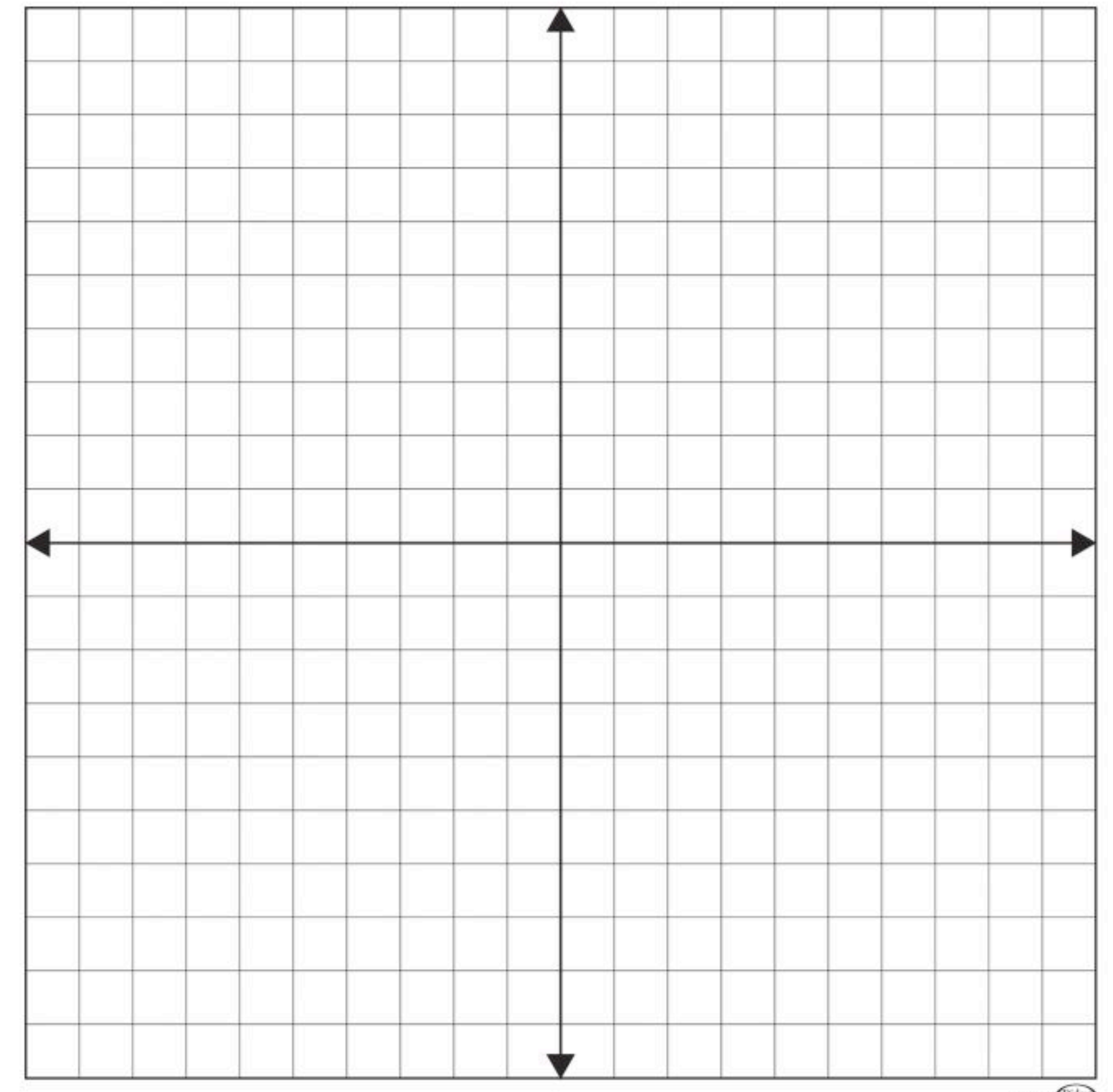
i j

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i j k

Identity Matrix as a Basis

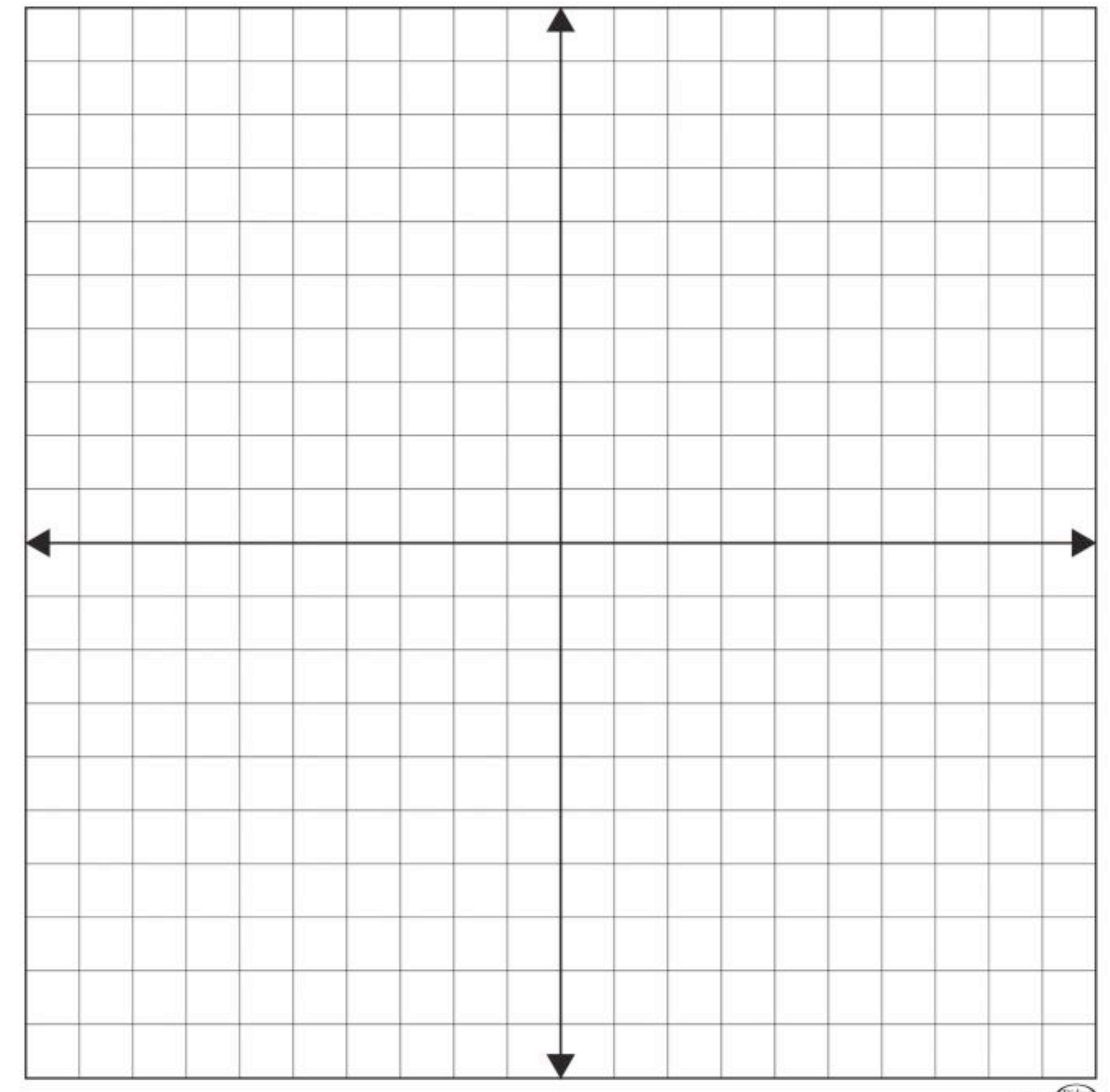
$$\begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Identity Matrix as a Basis

- Vectors can be viewed as being **composed of the Standard Basis vectors**

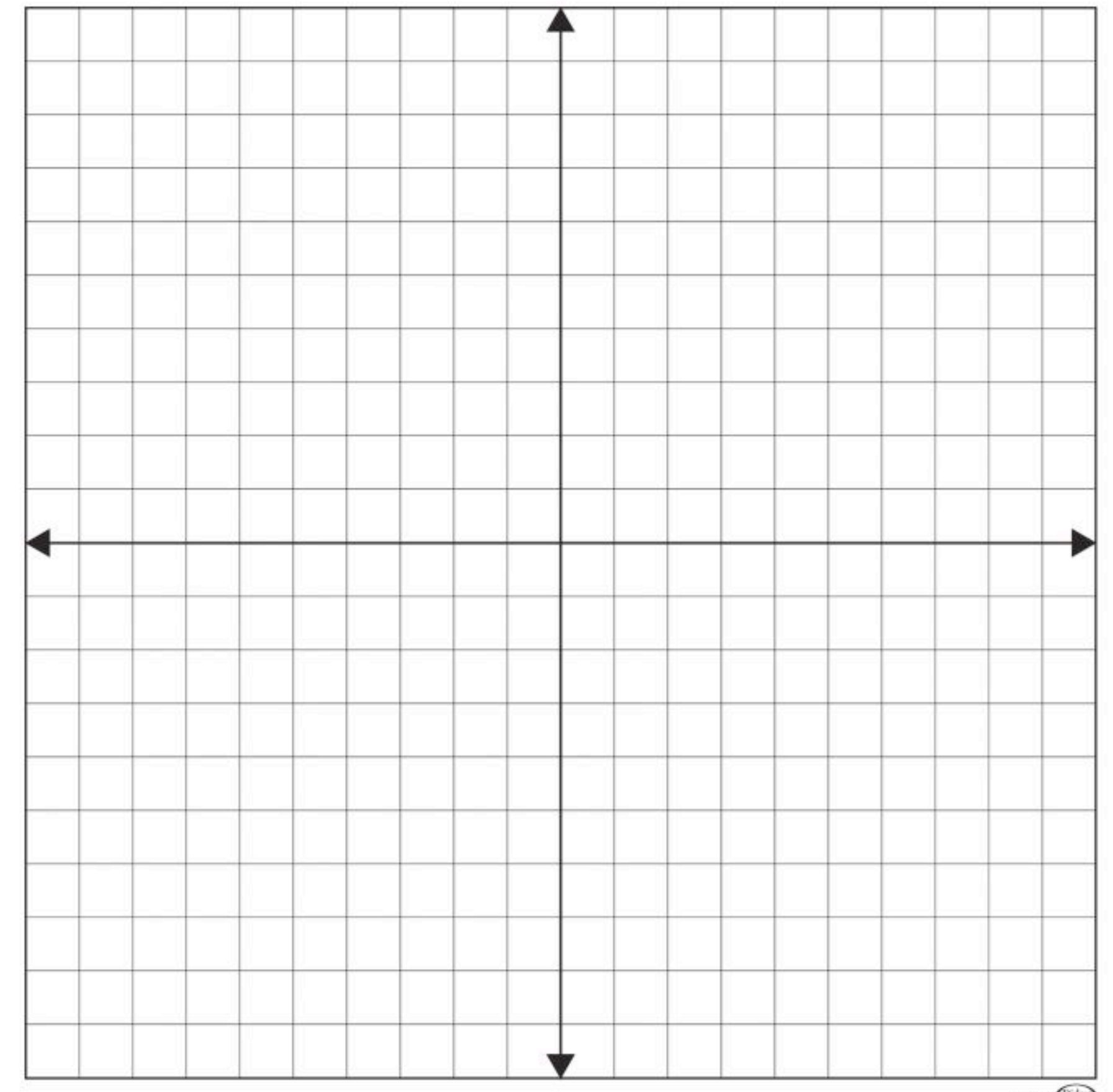
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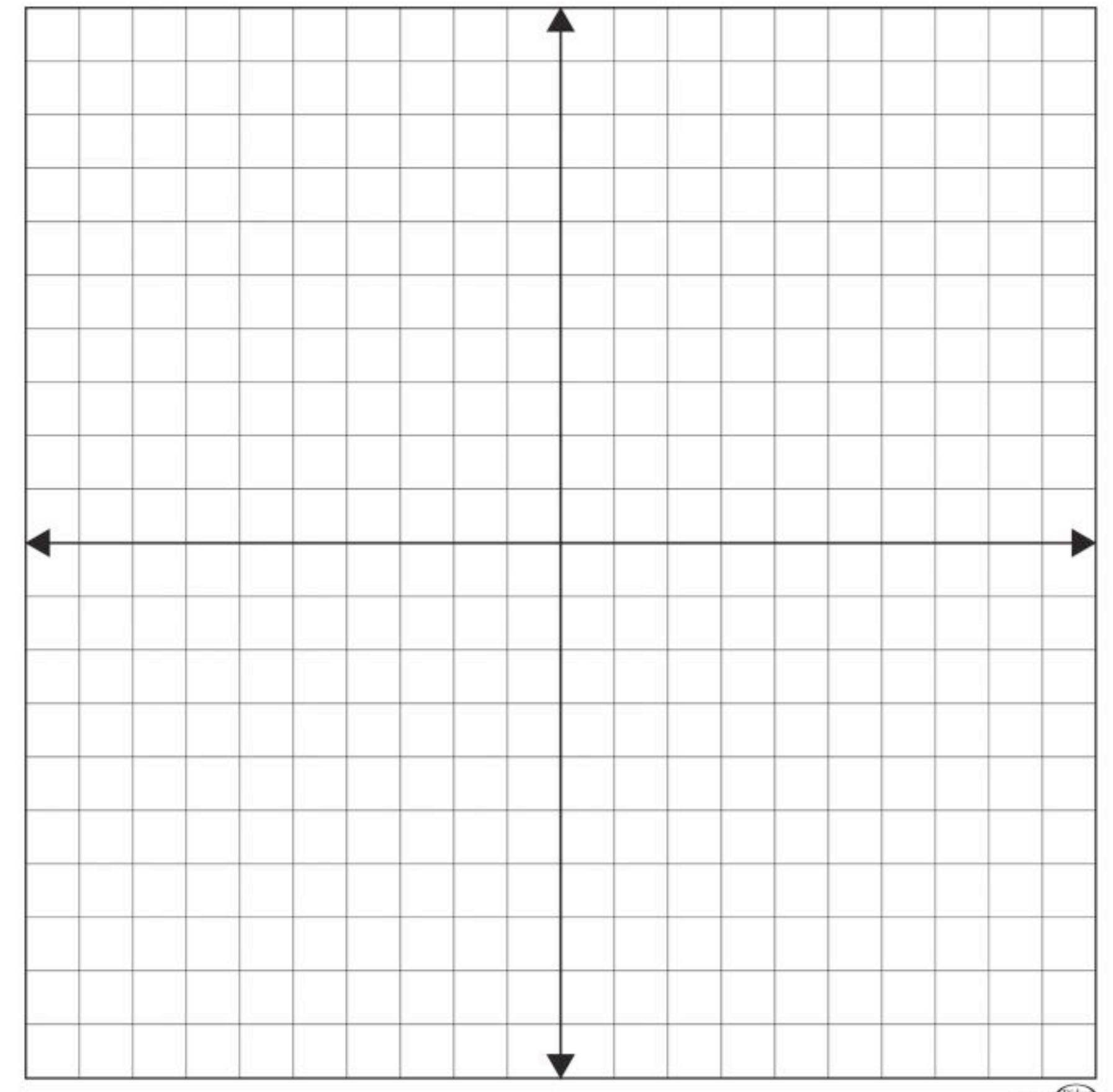
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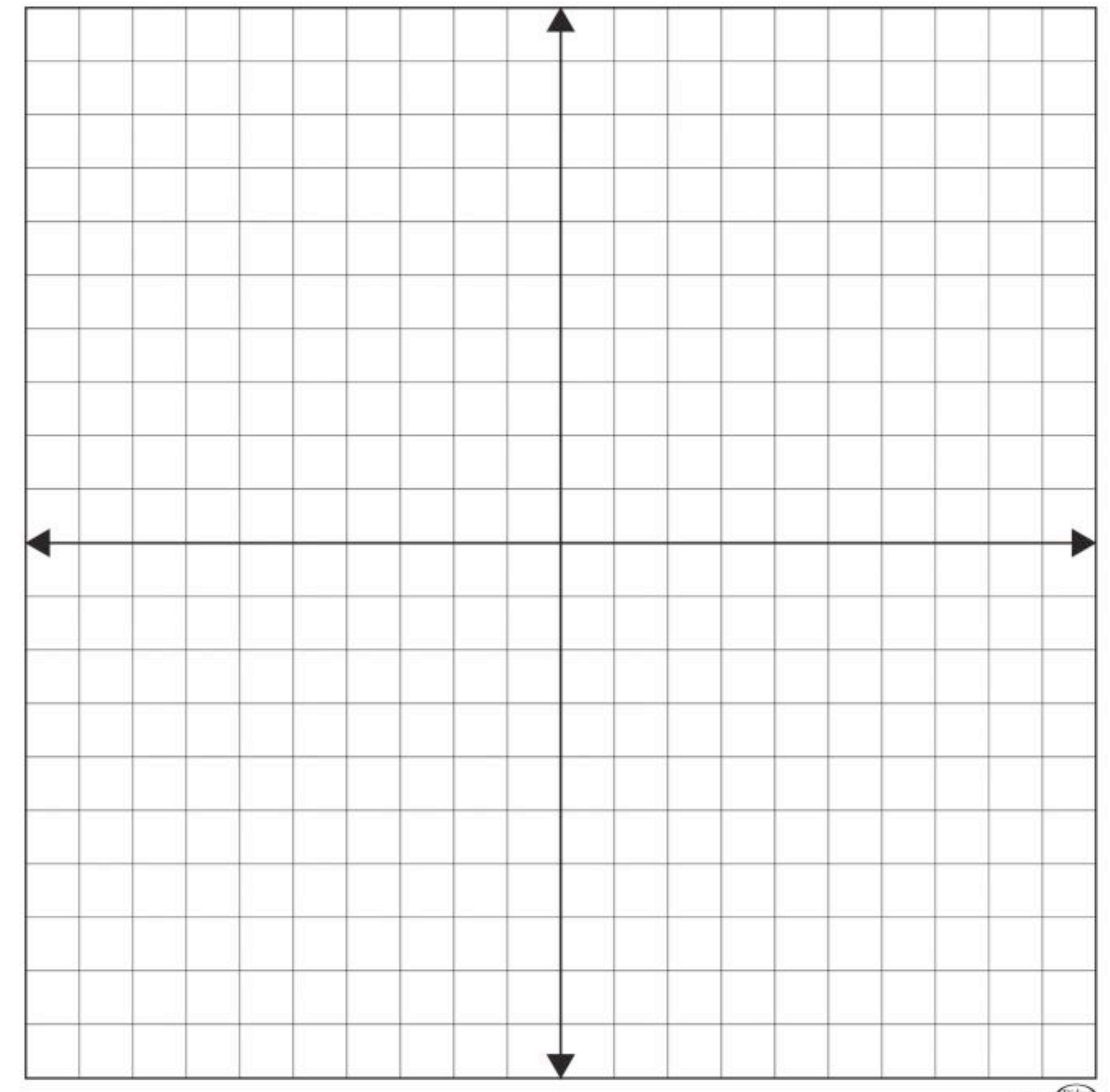
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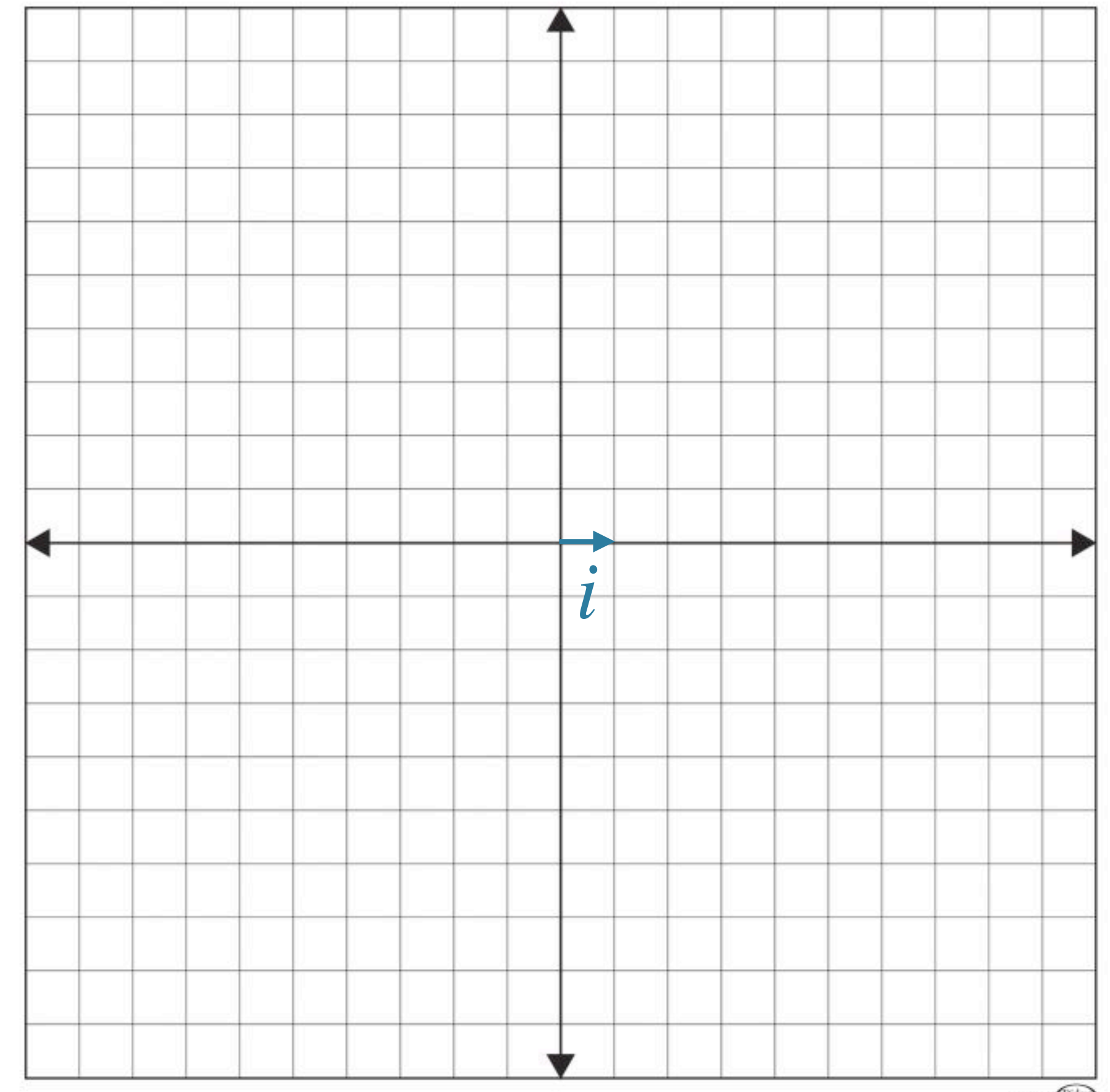
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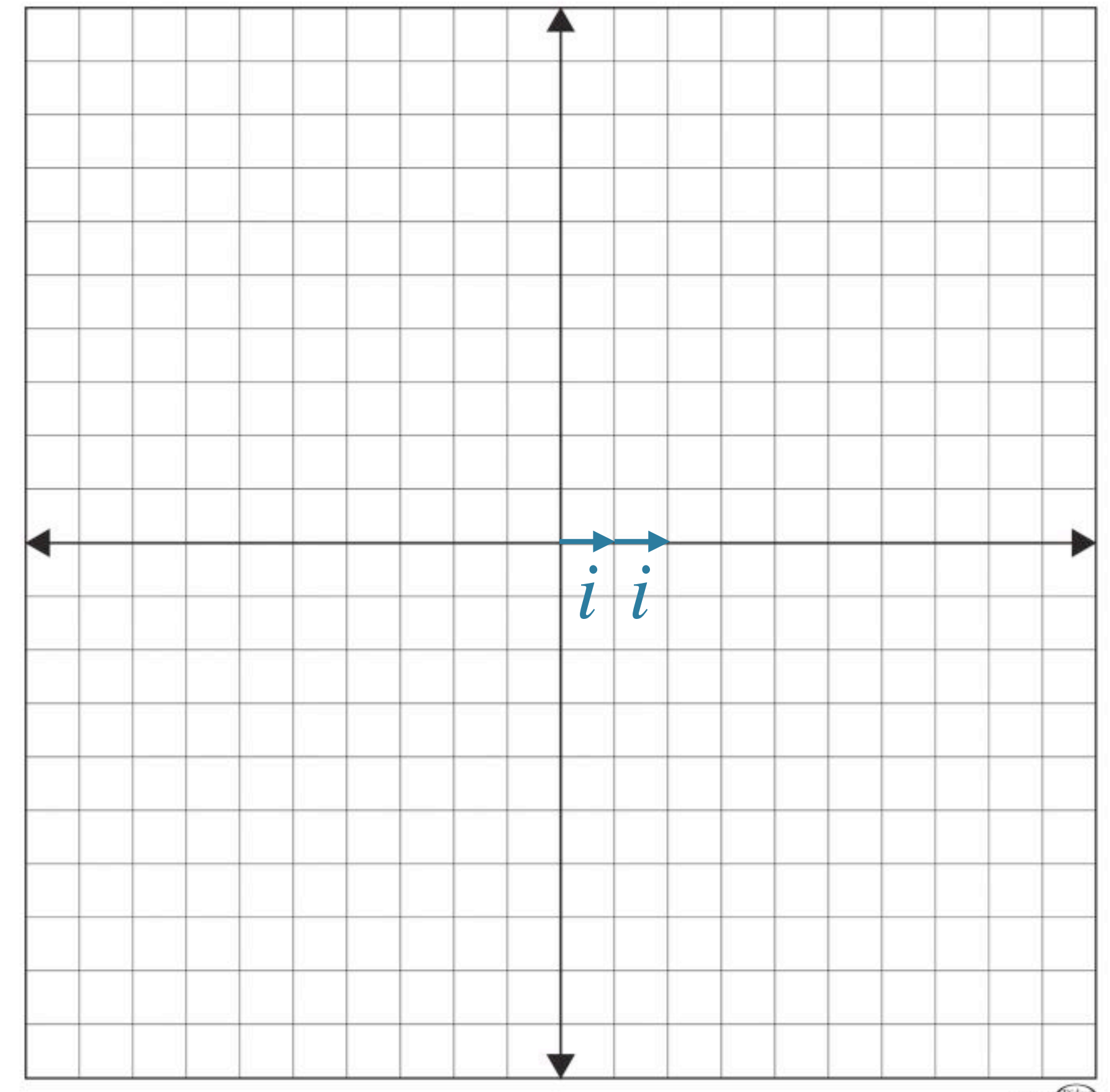
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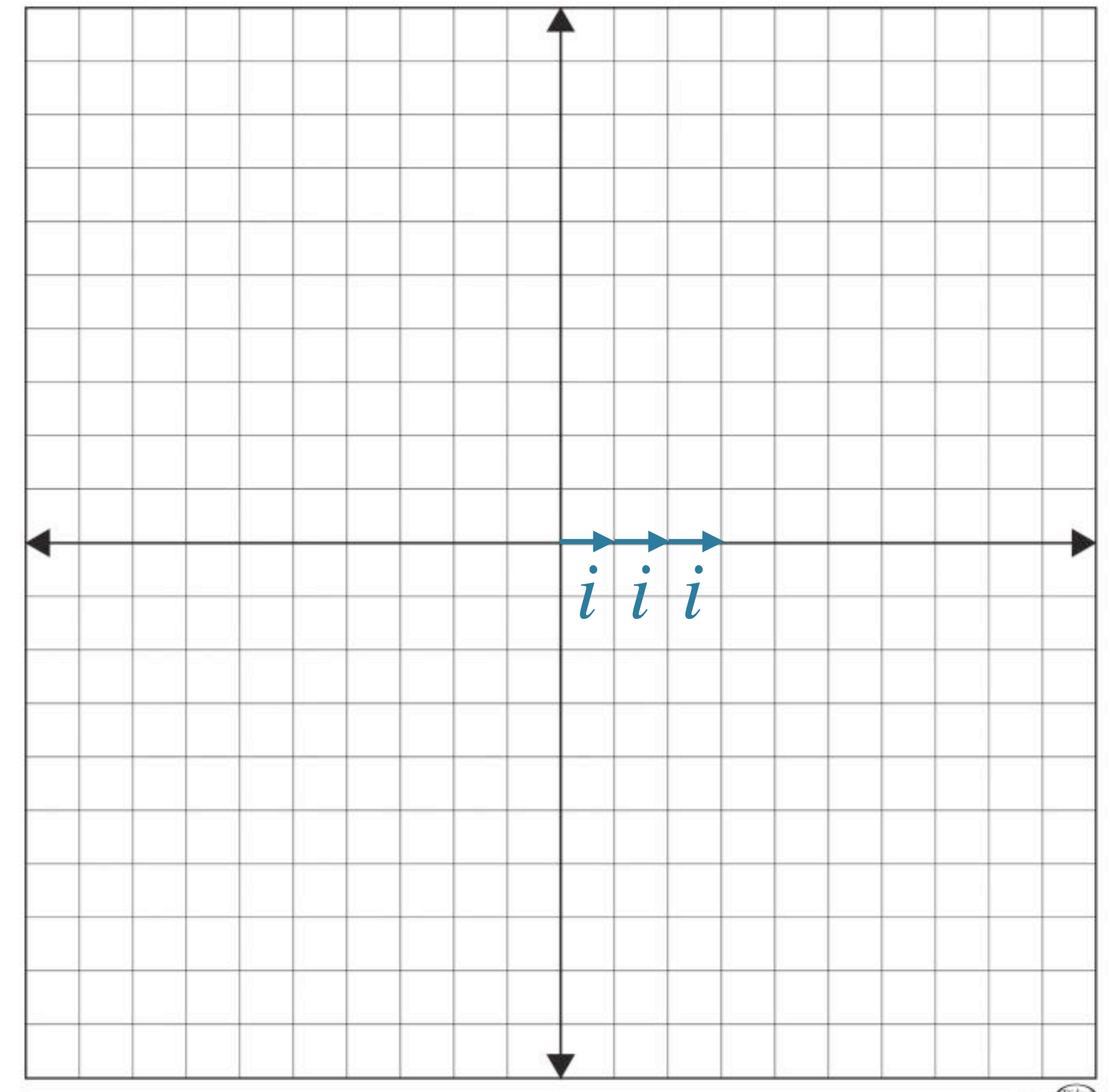
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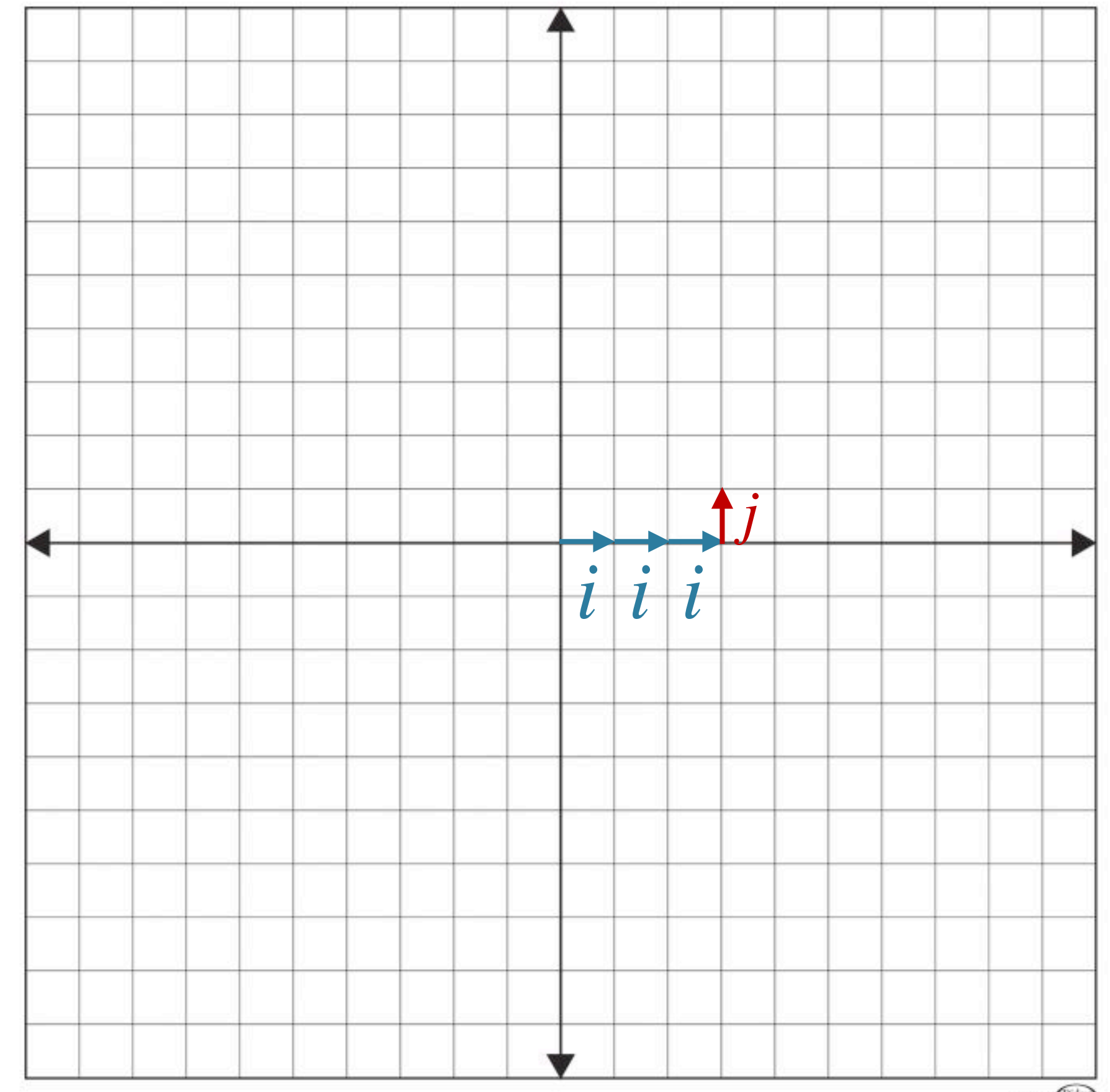
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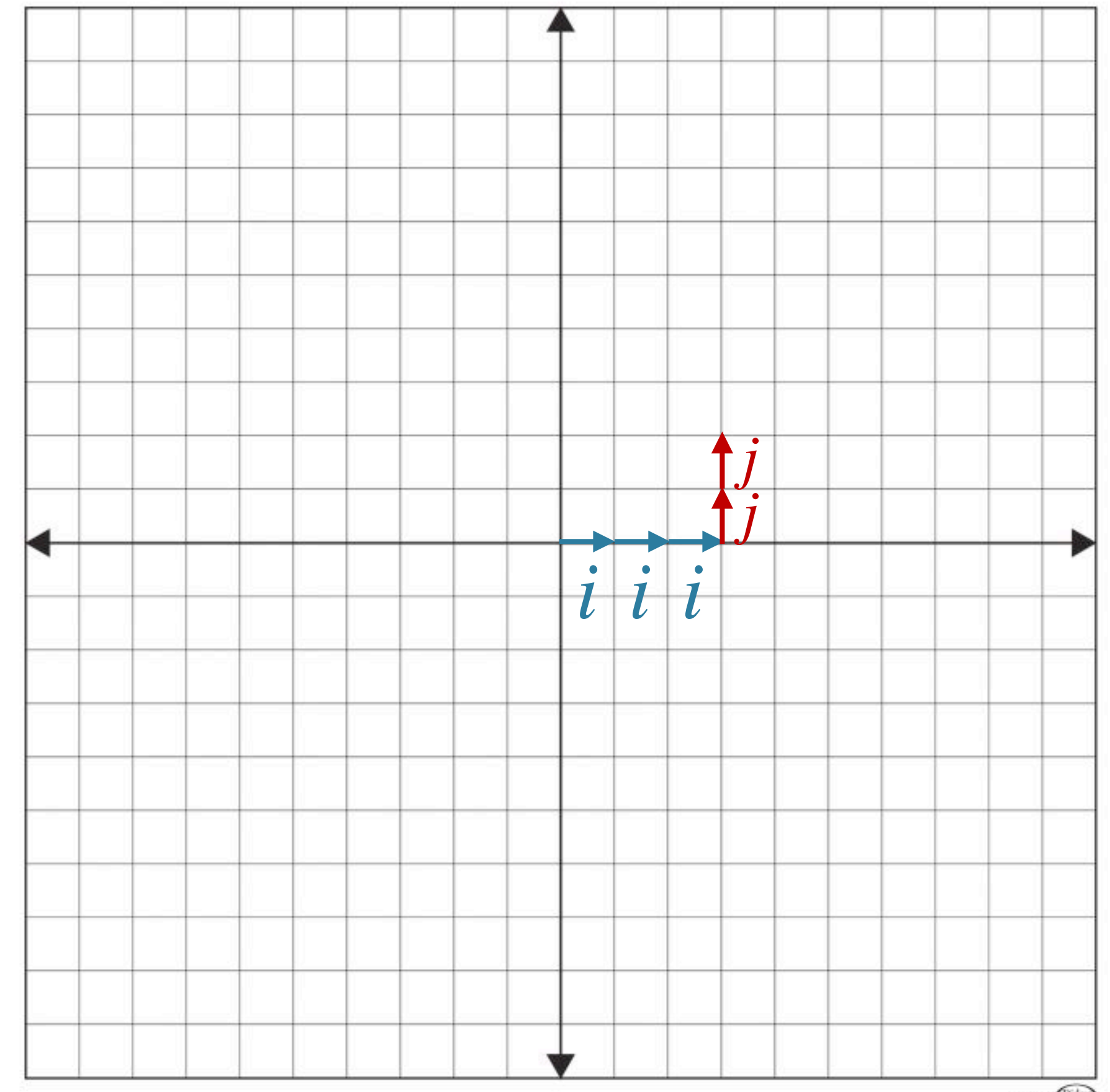
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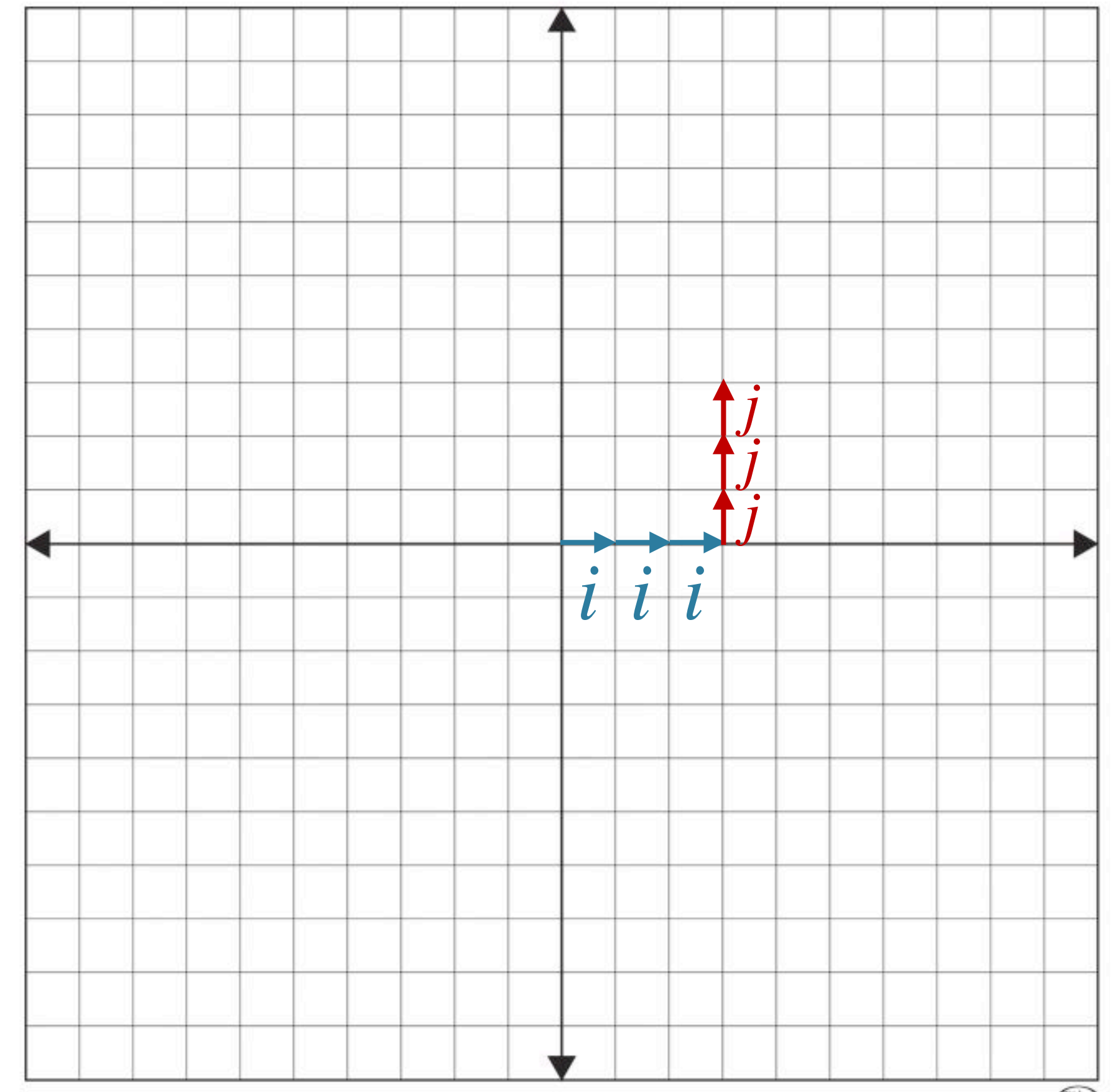
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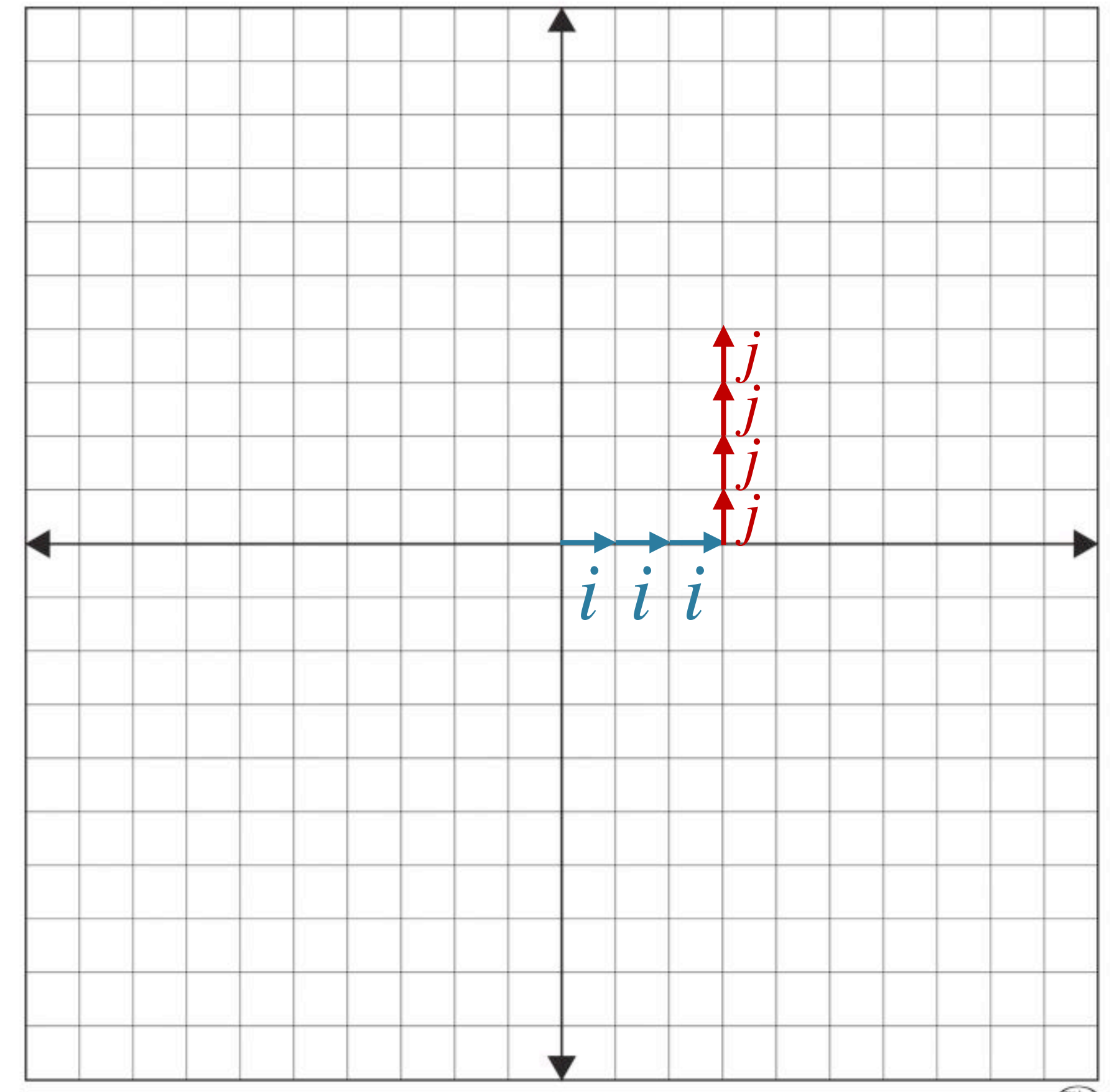
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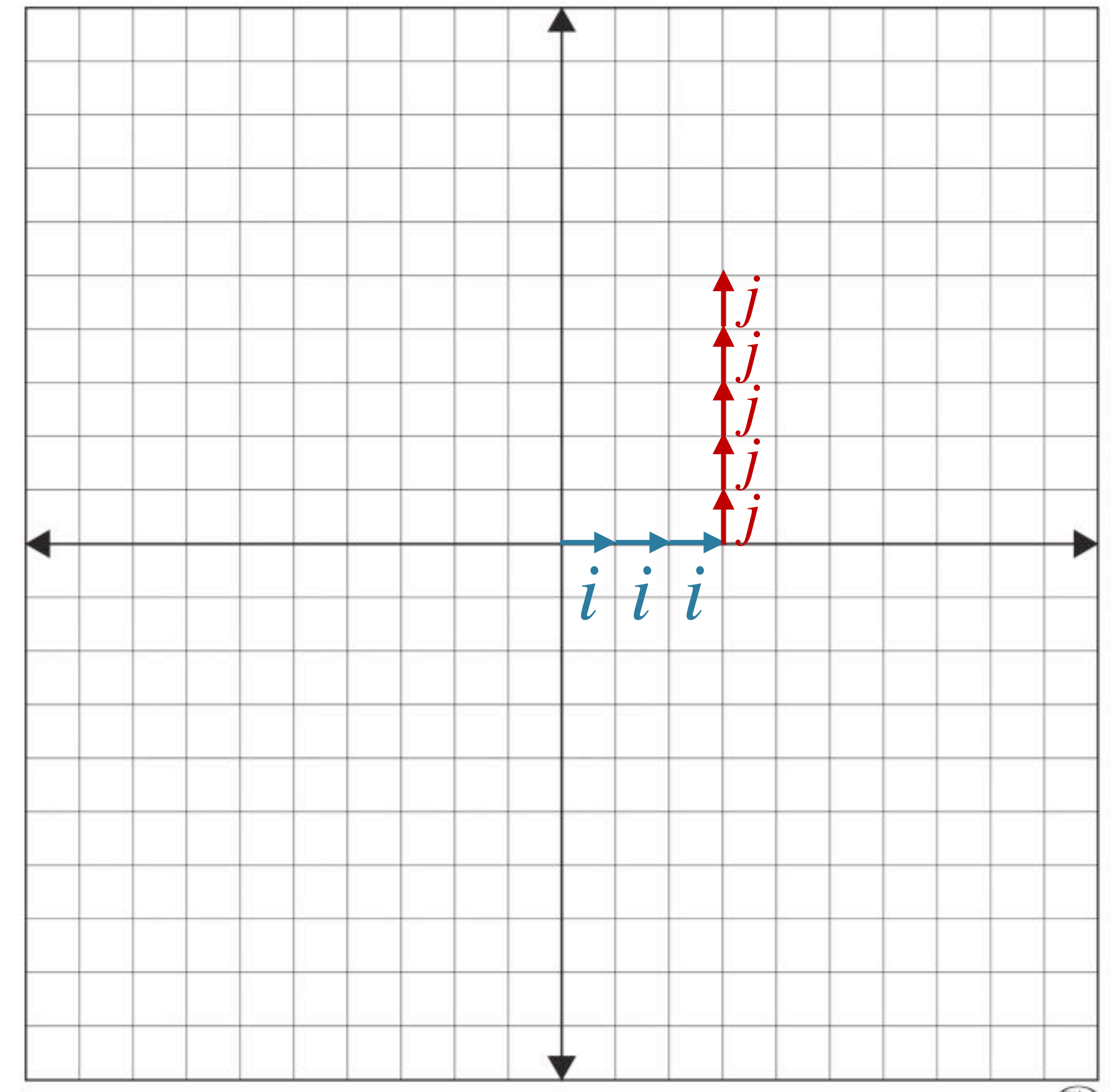
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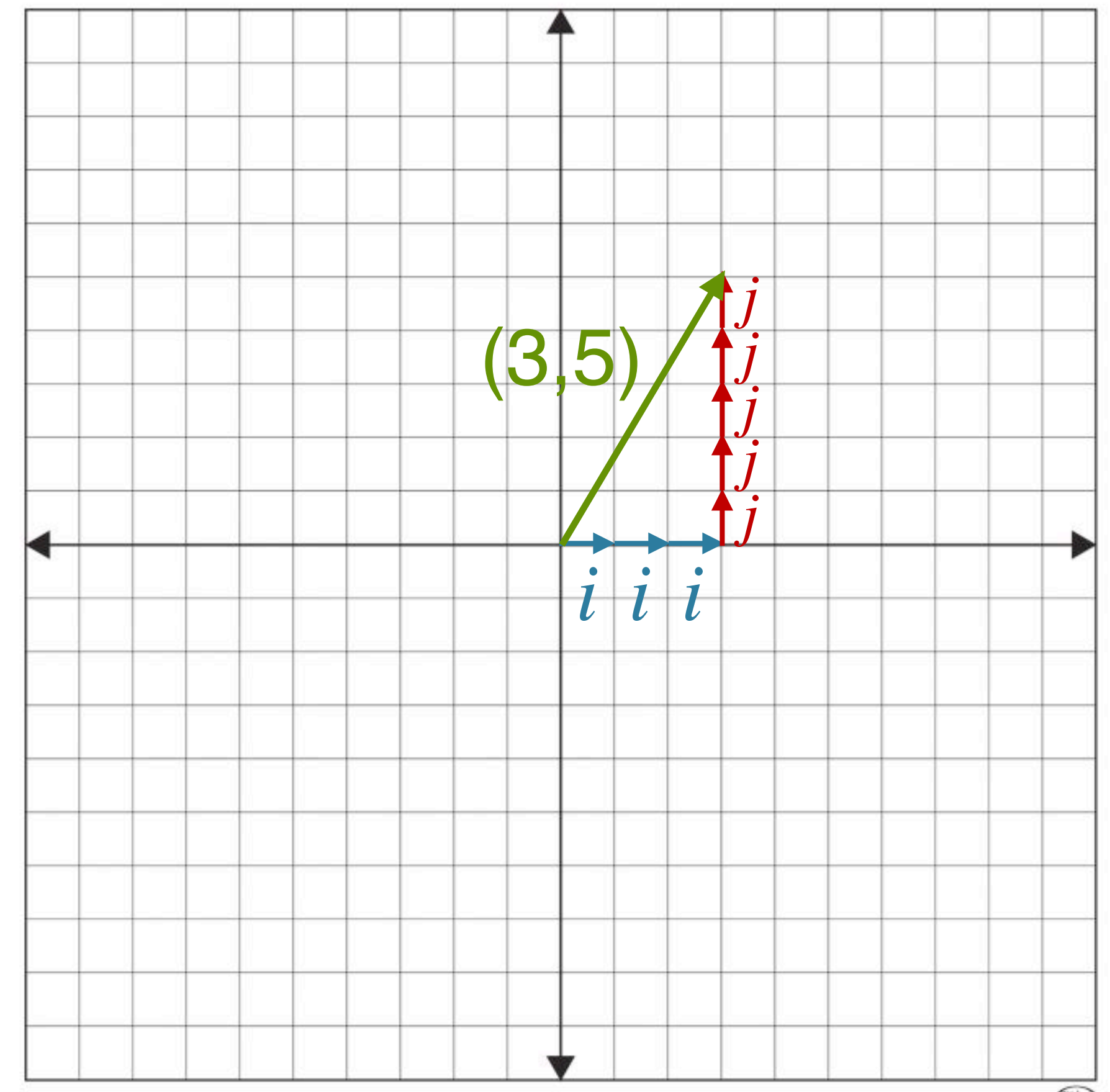
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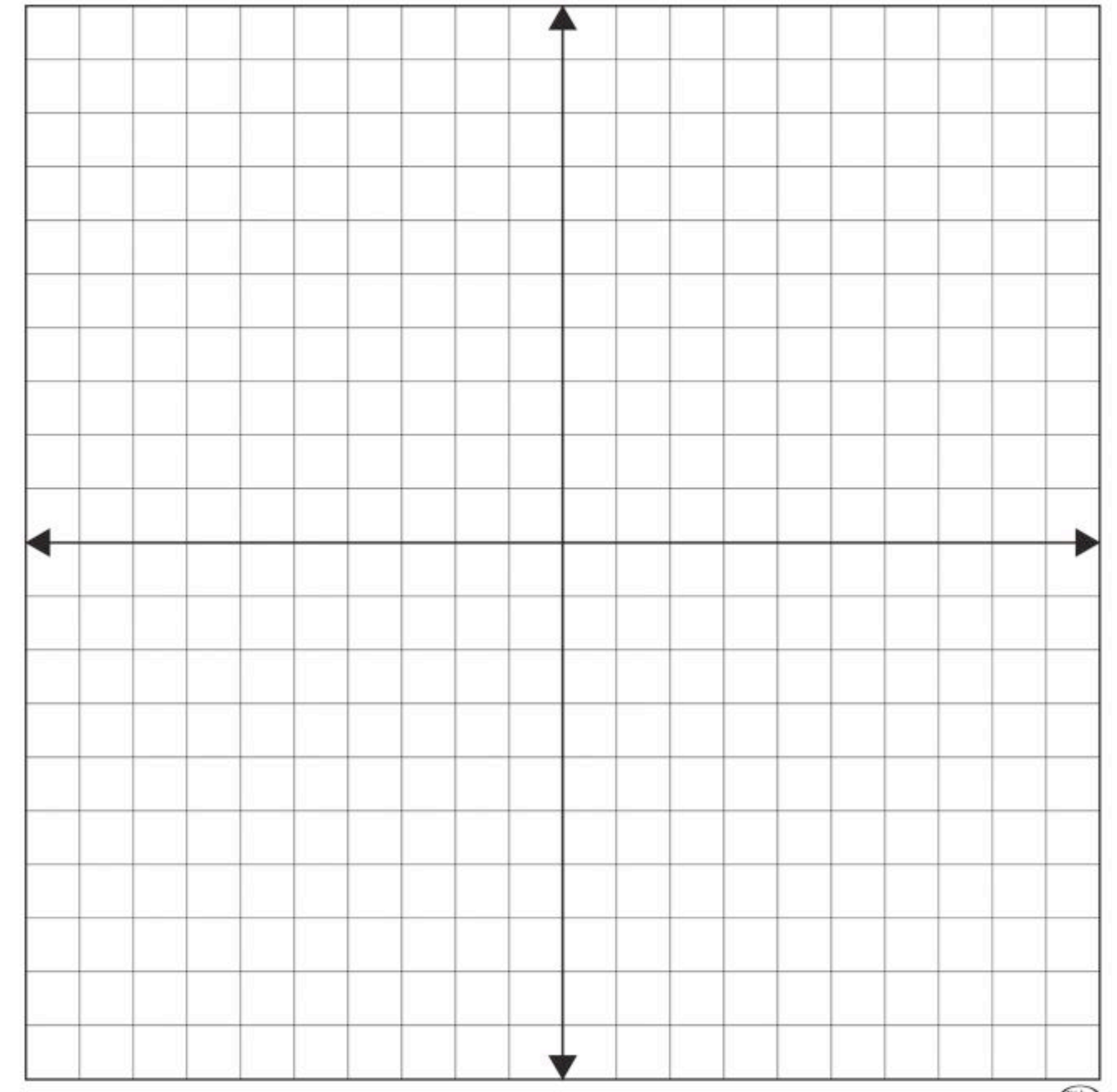
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Linear Transformation

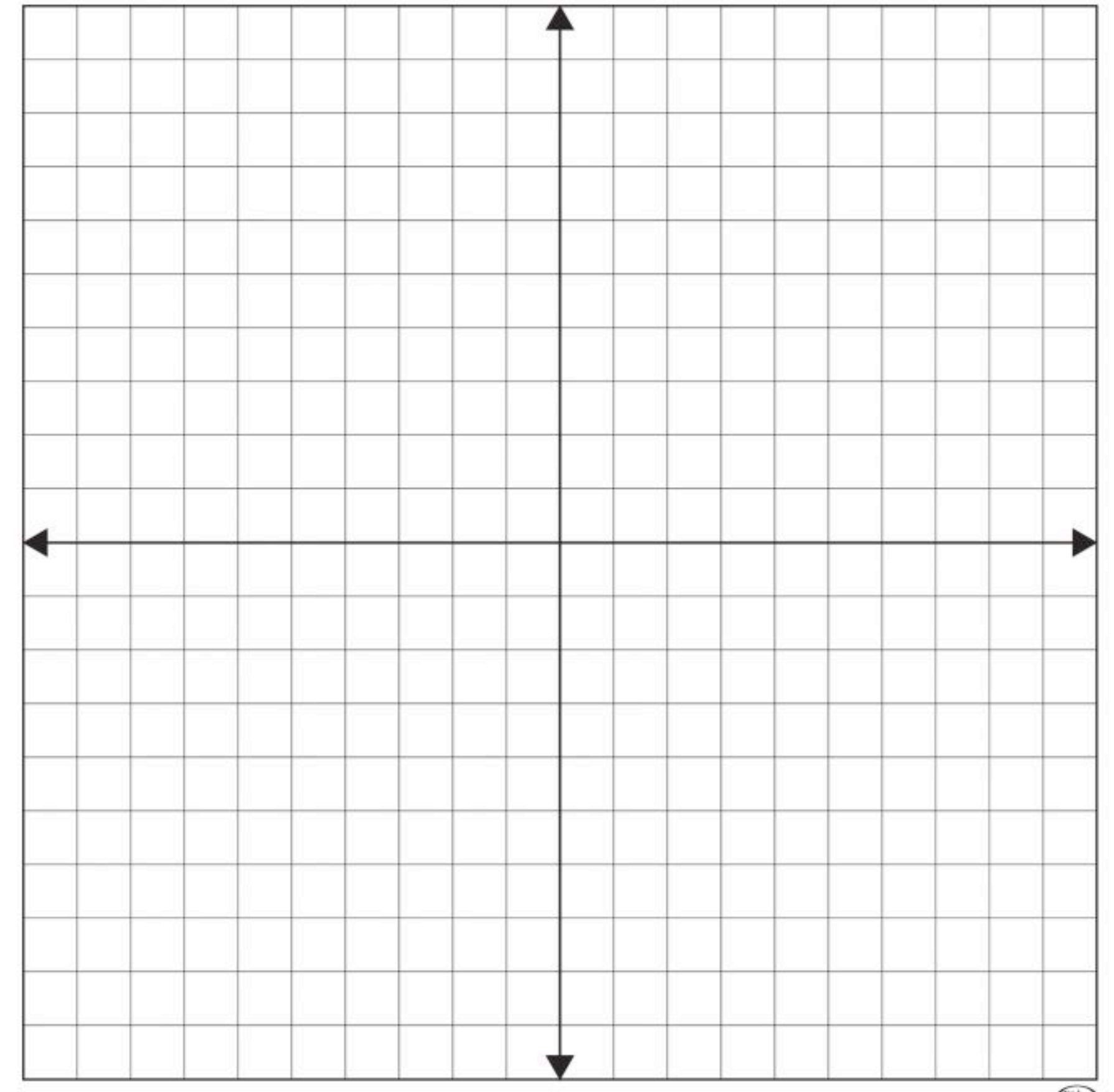
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Linear Transformation

- Multiplying by a matrix converts a vector to a **new basis**

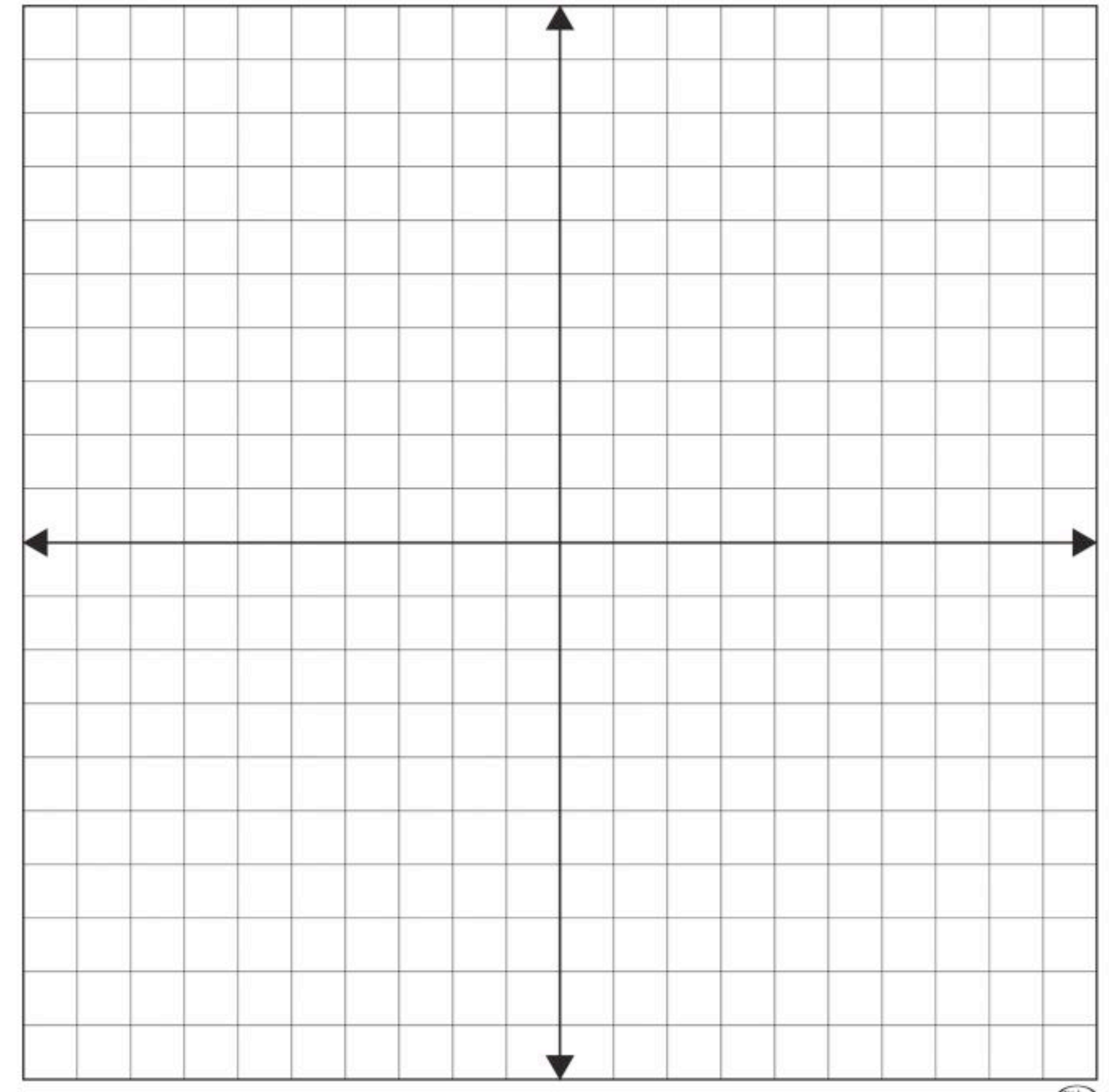
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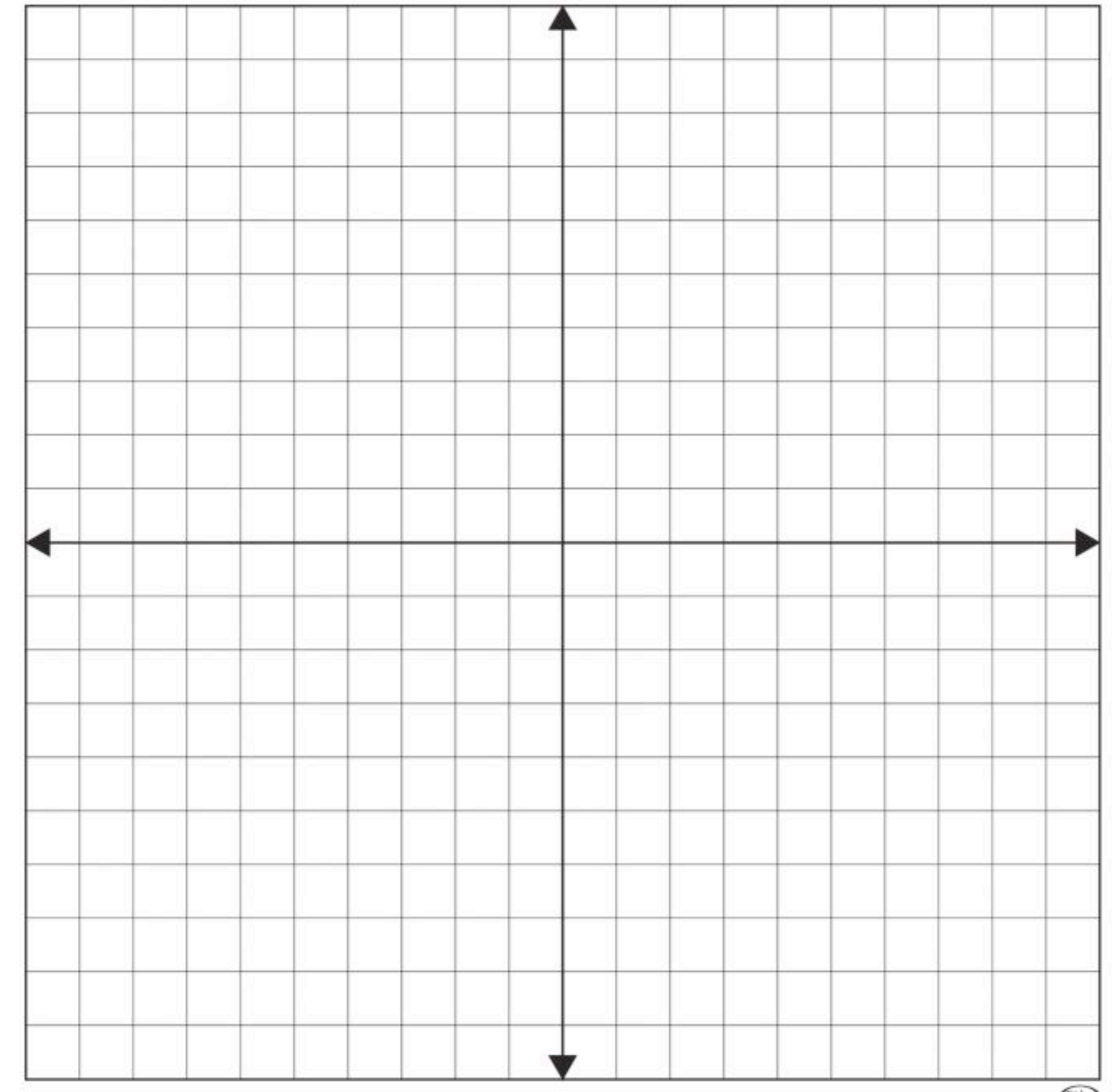


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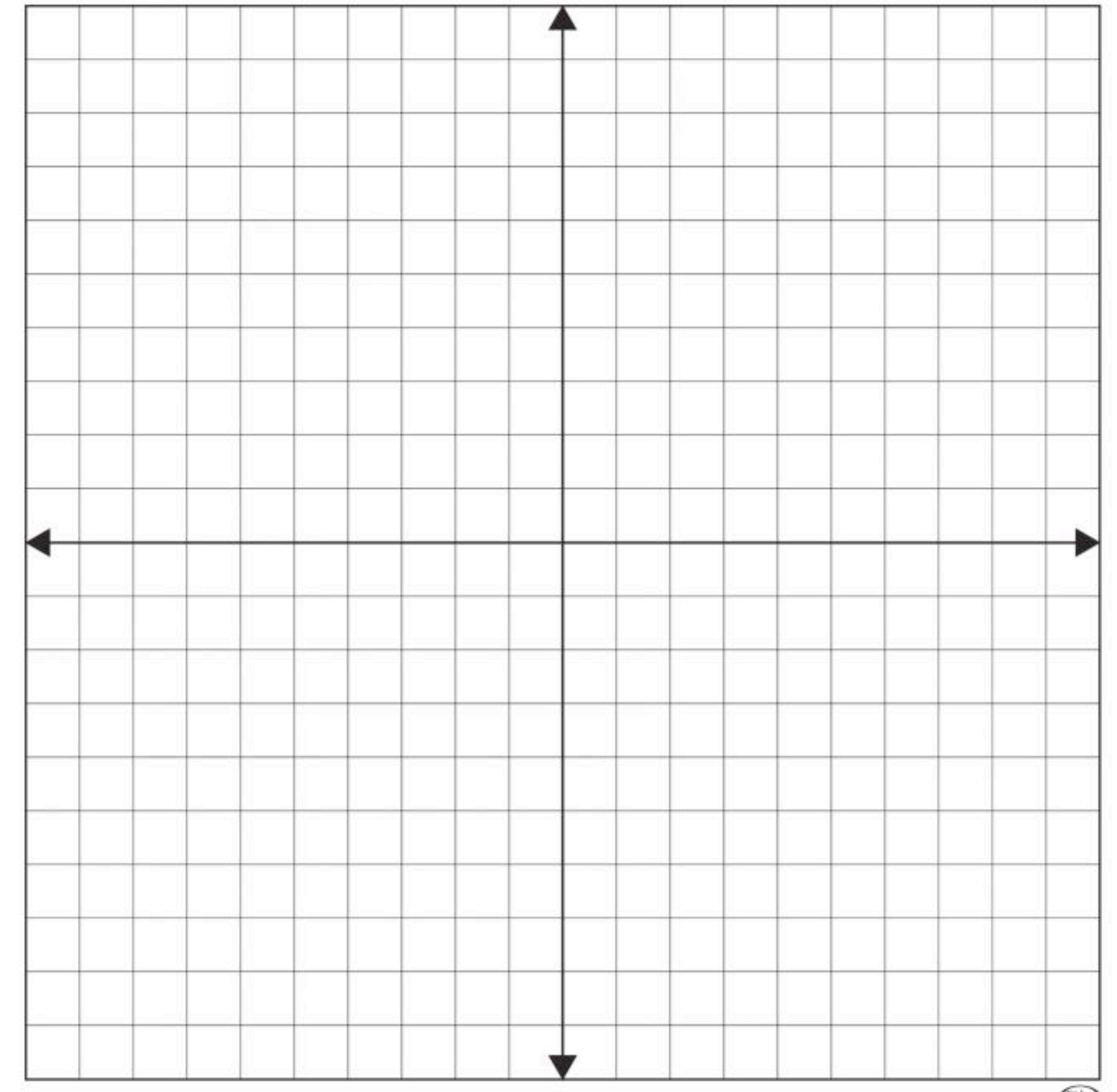


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new basis i'

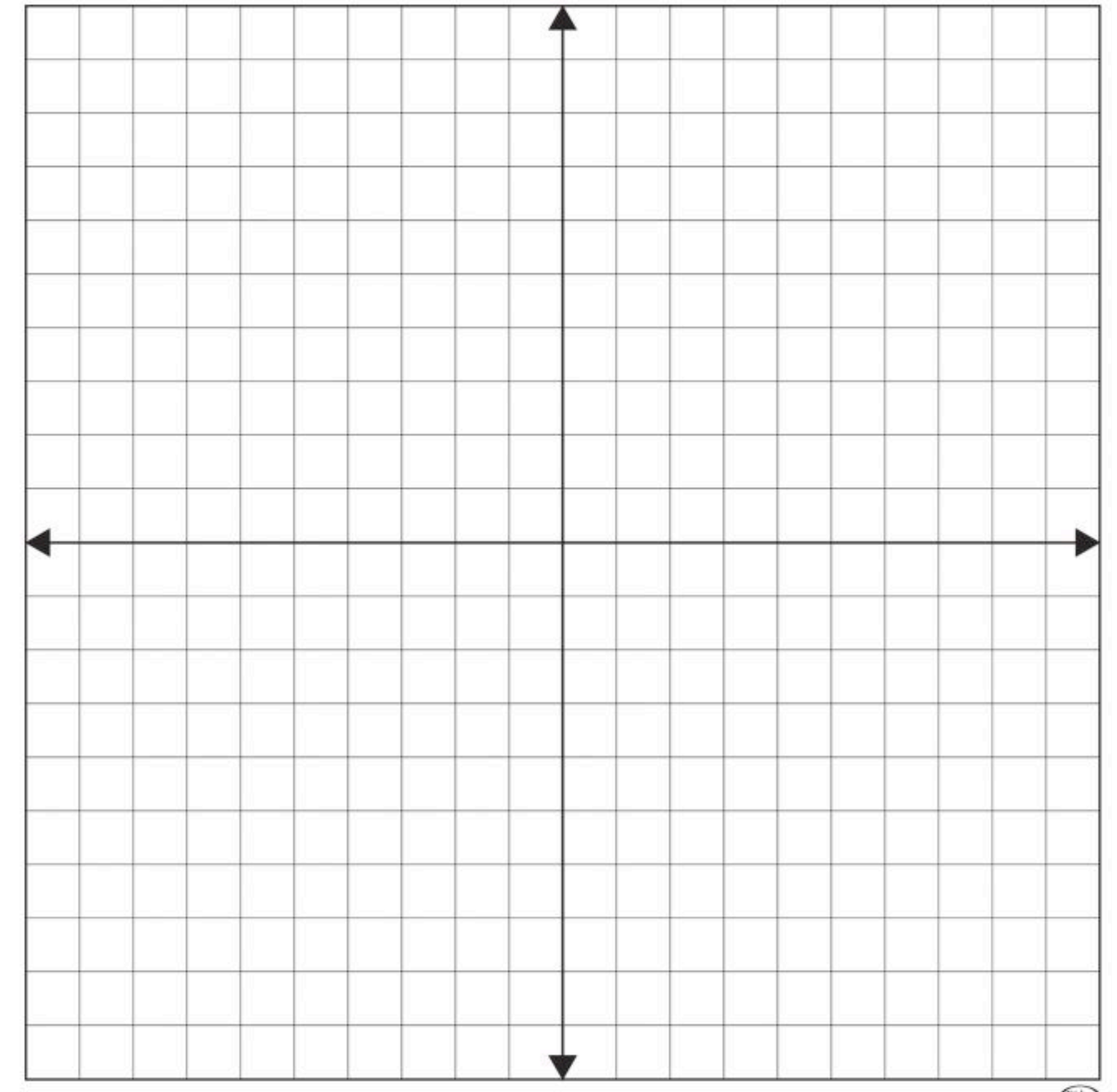


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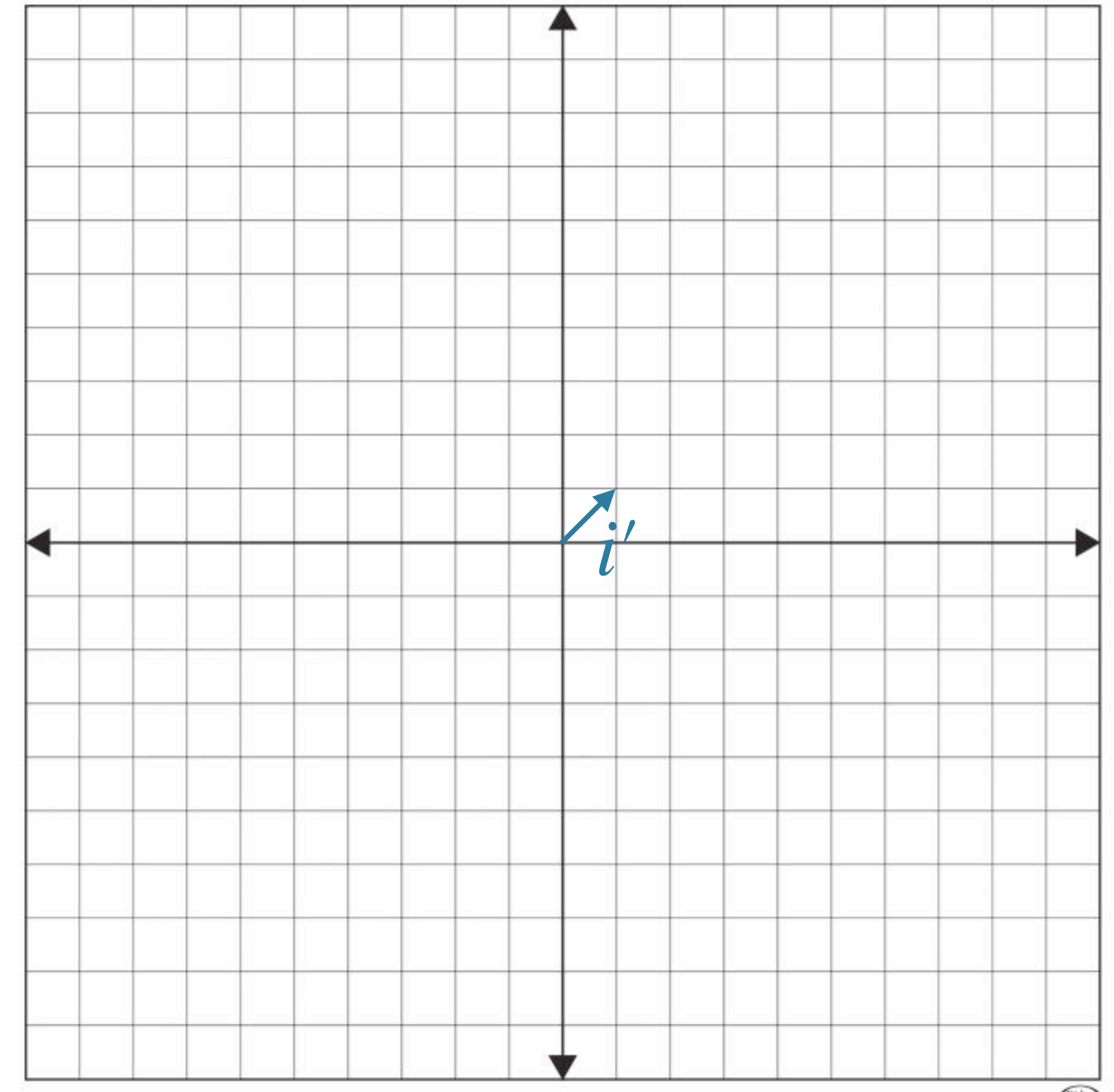


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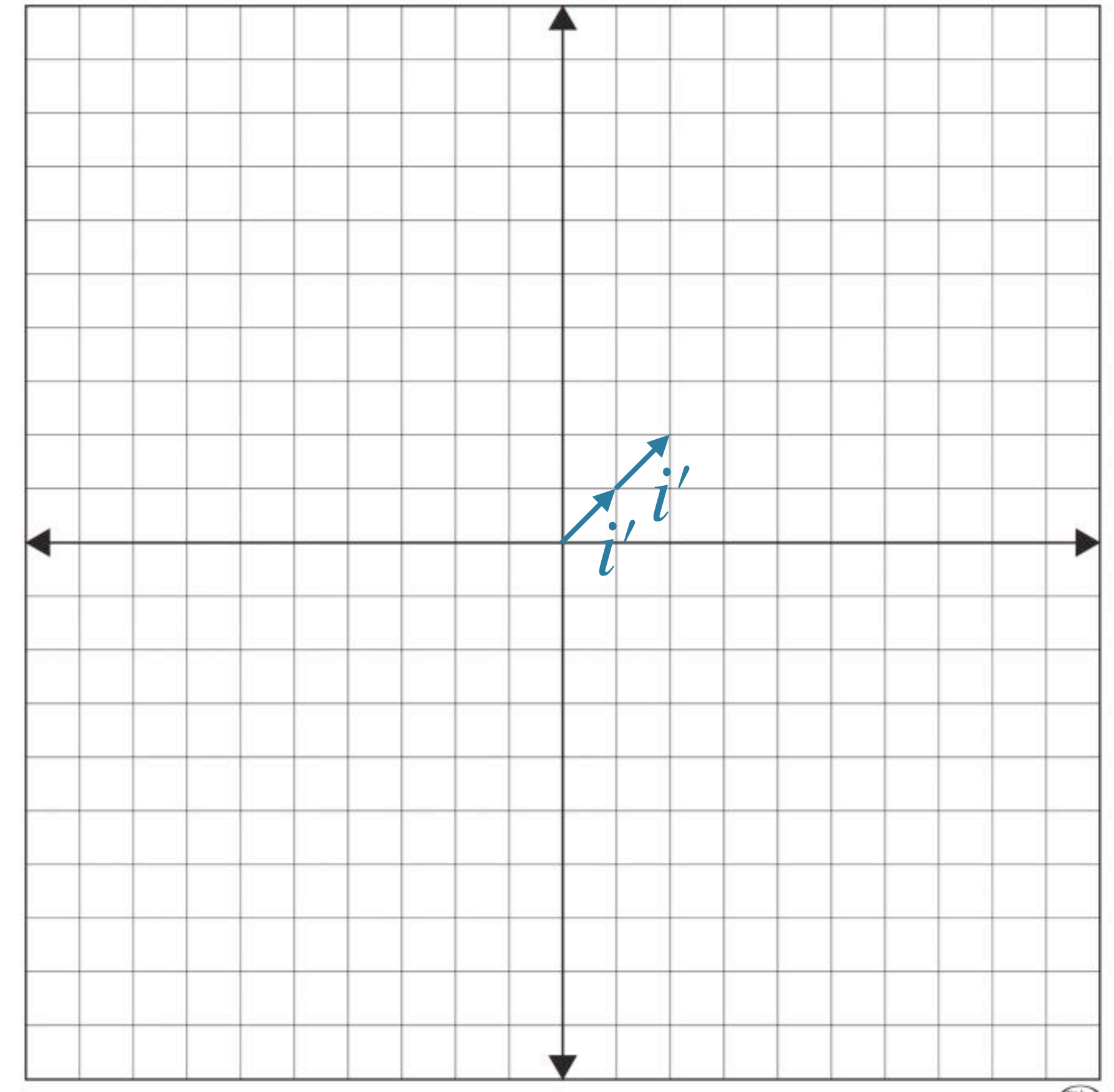


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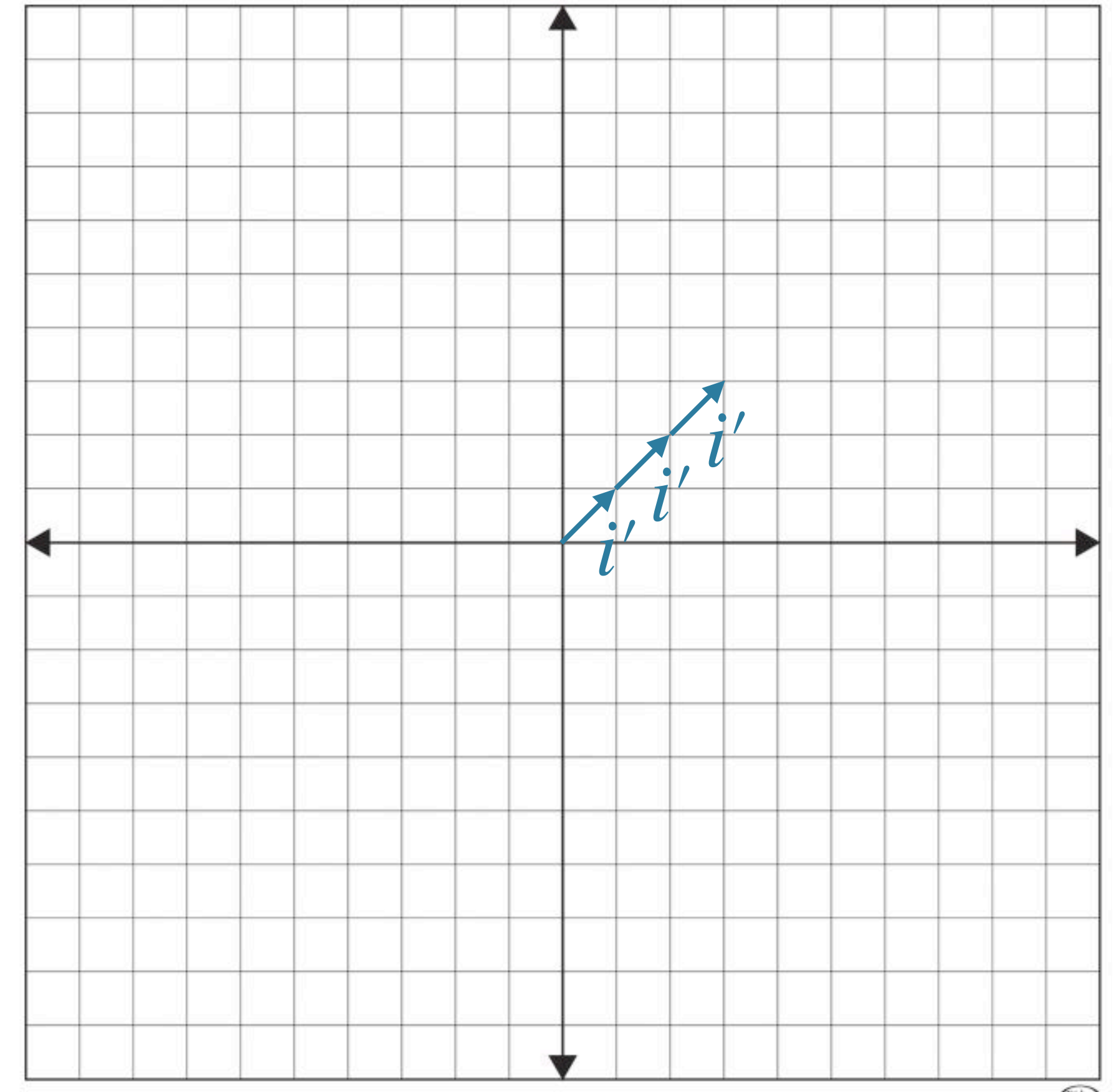


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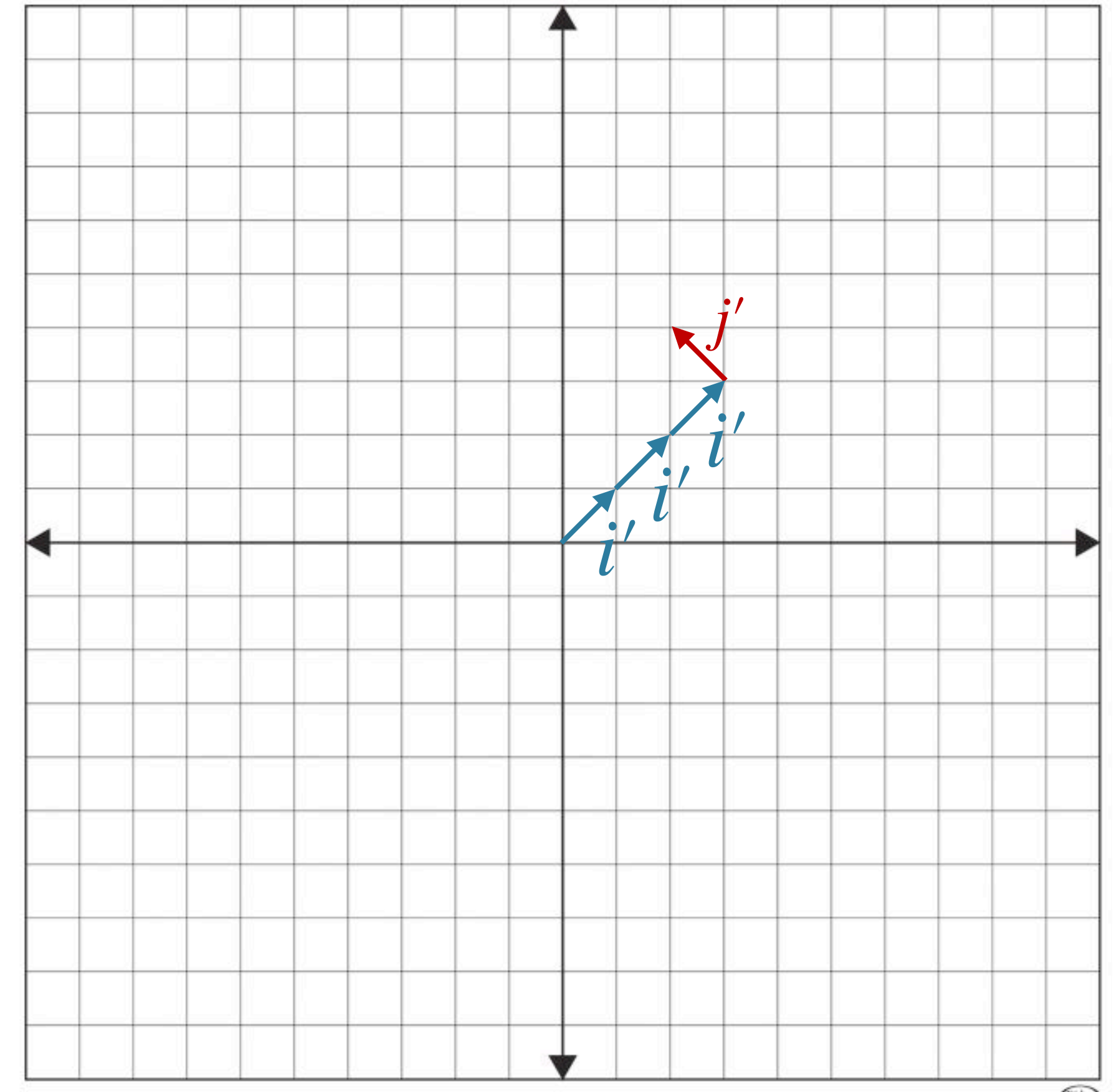


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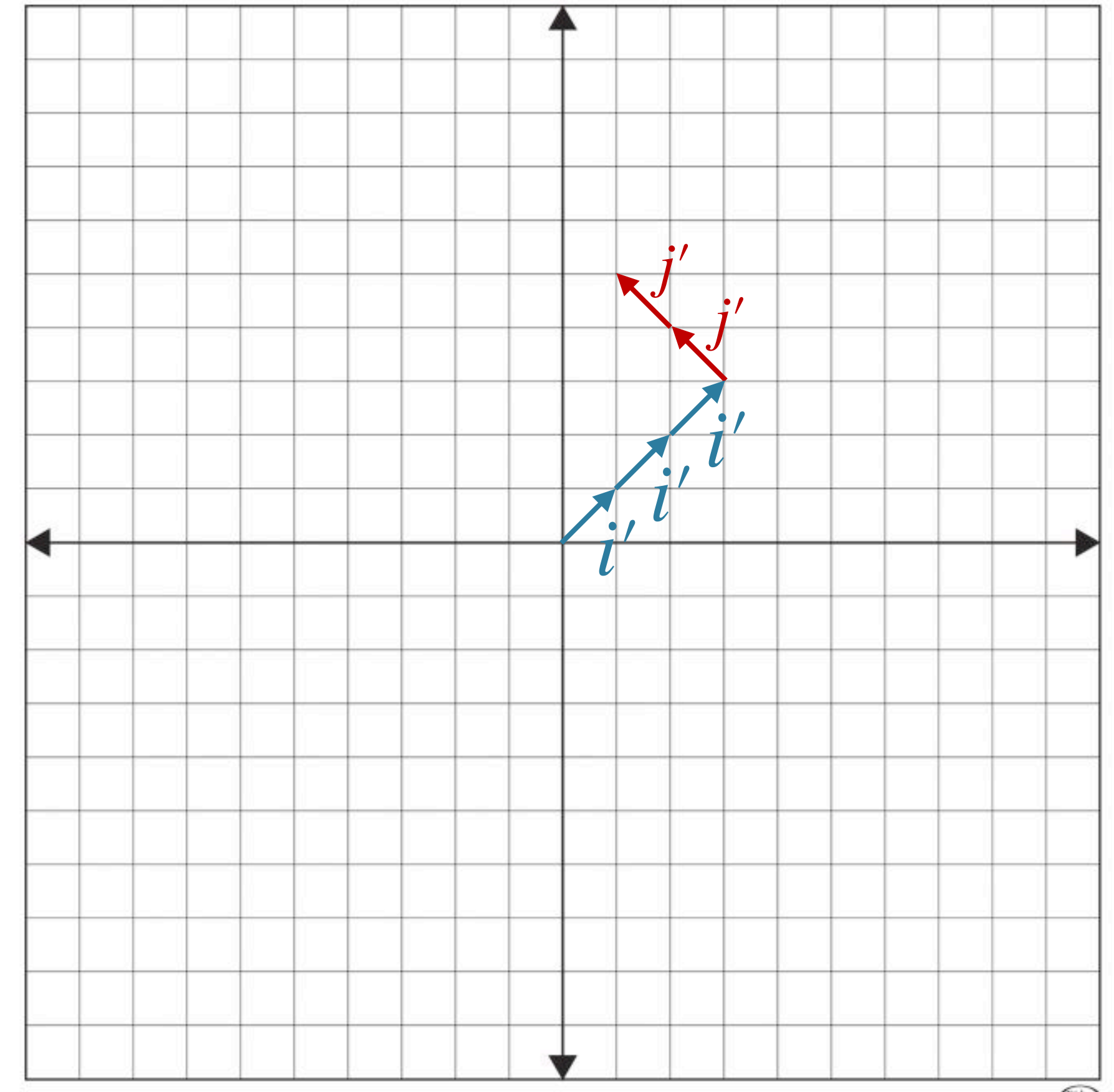


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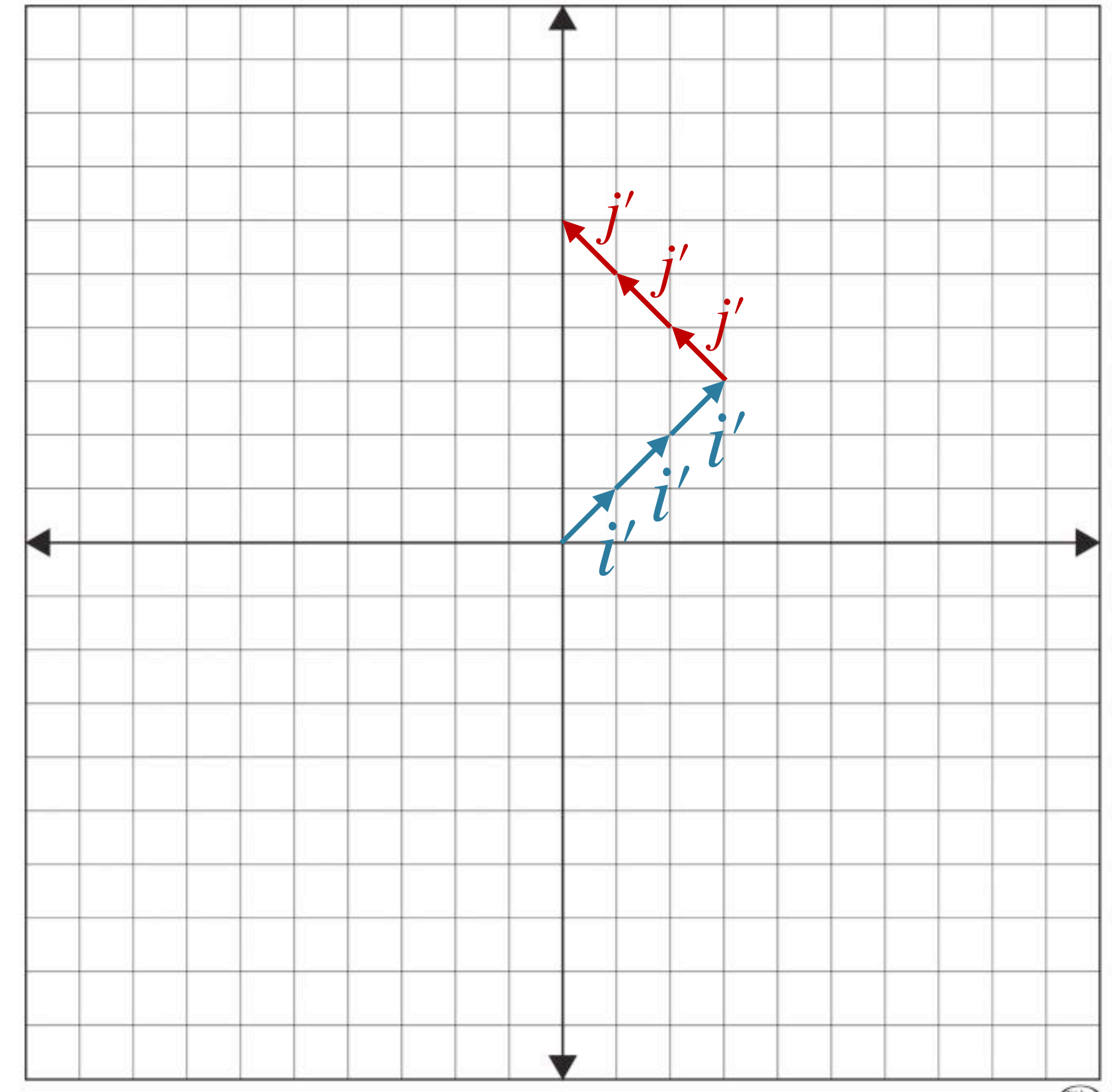


Linear Transformation

- Multiplying by a matrix converts a vector to a **new basis**
 - The basis consists of the **matrix columns**

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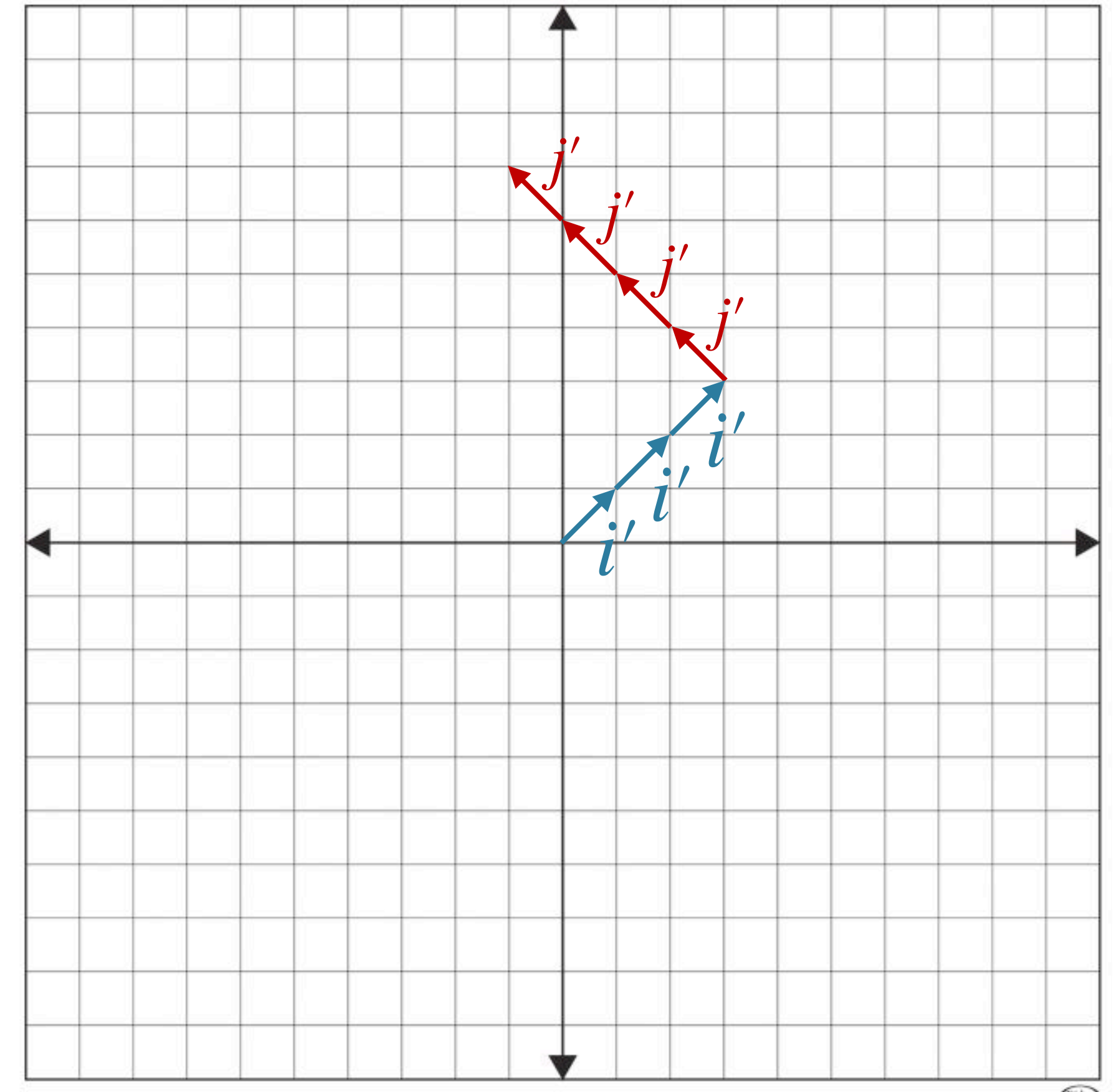


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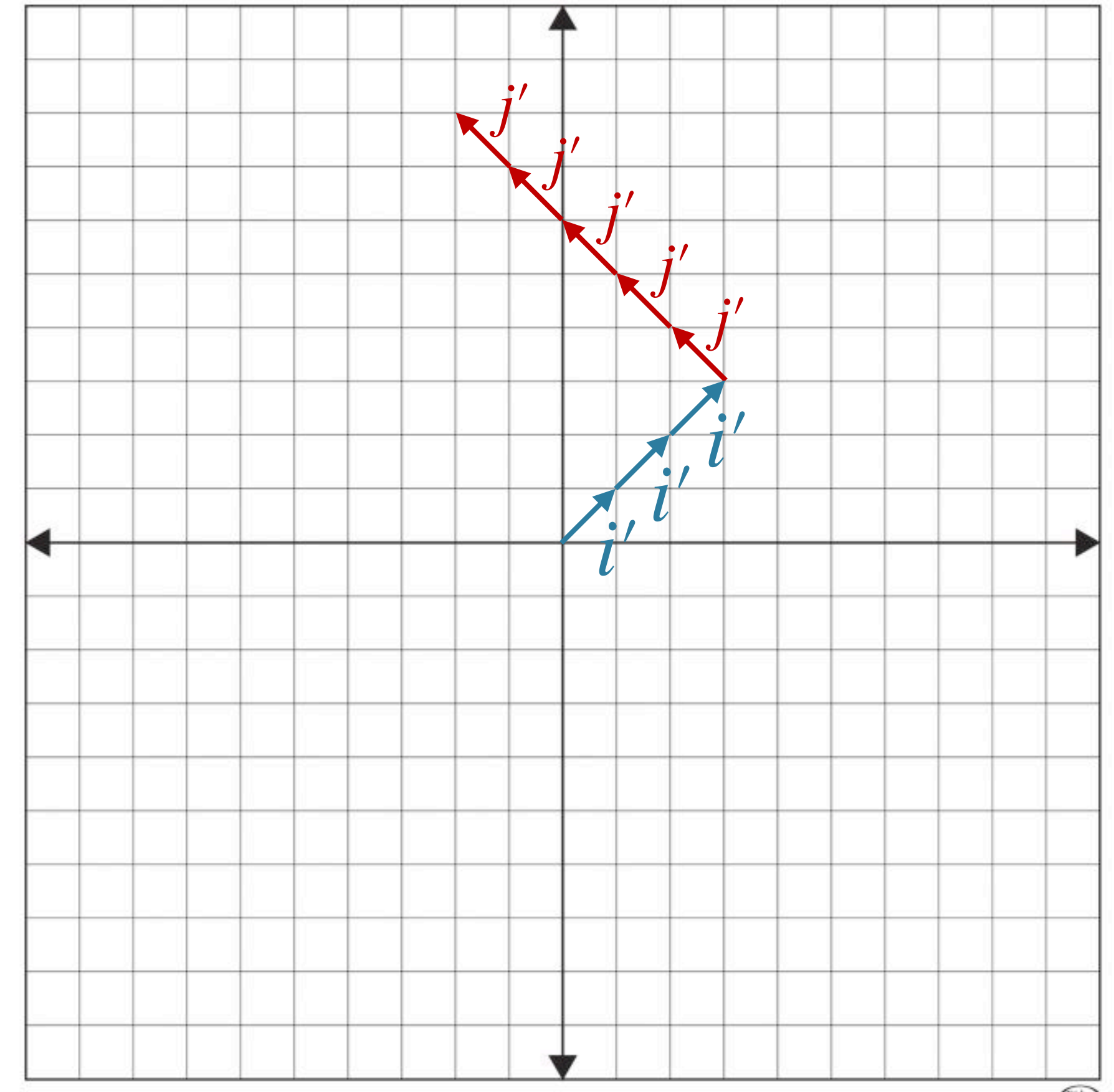


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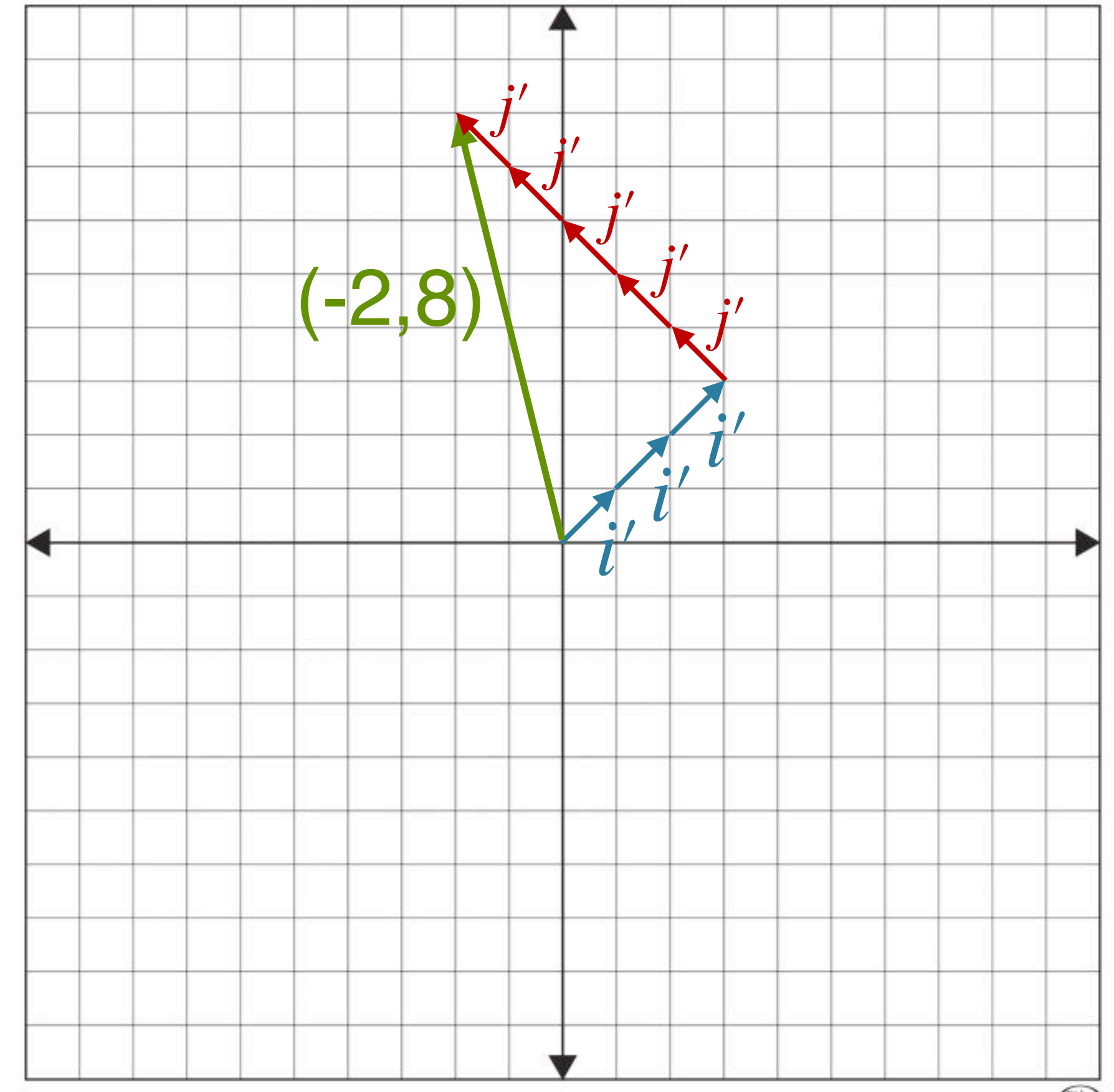


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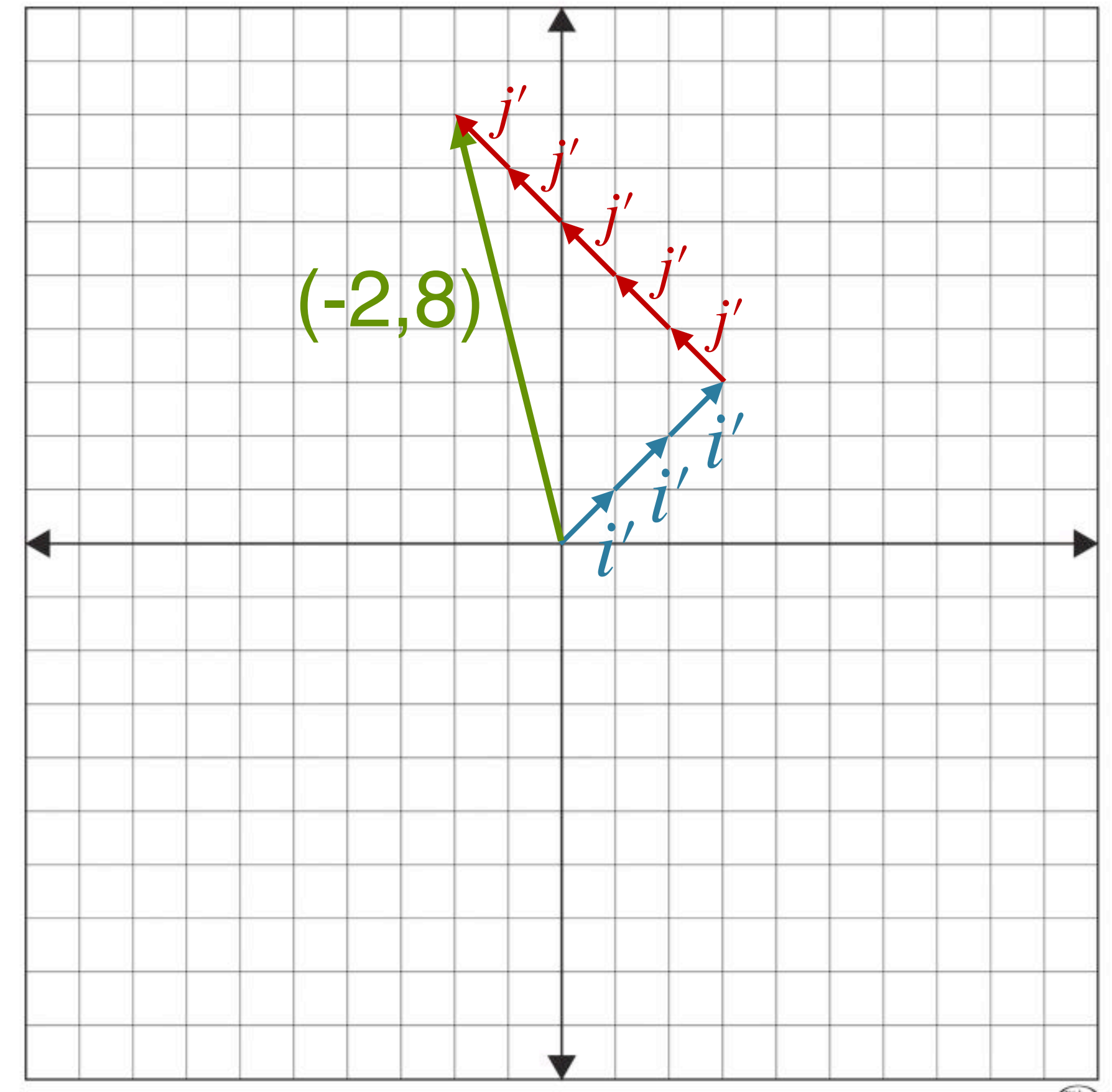


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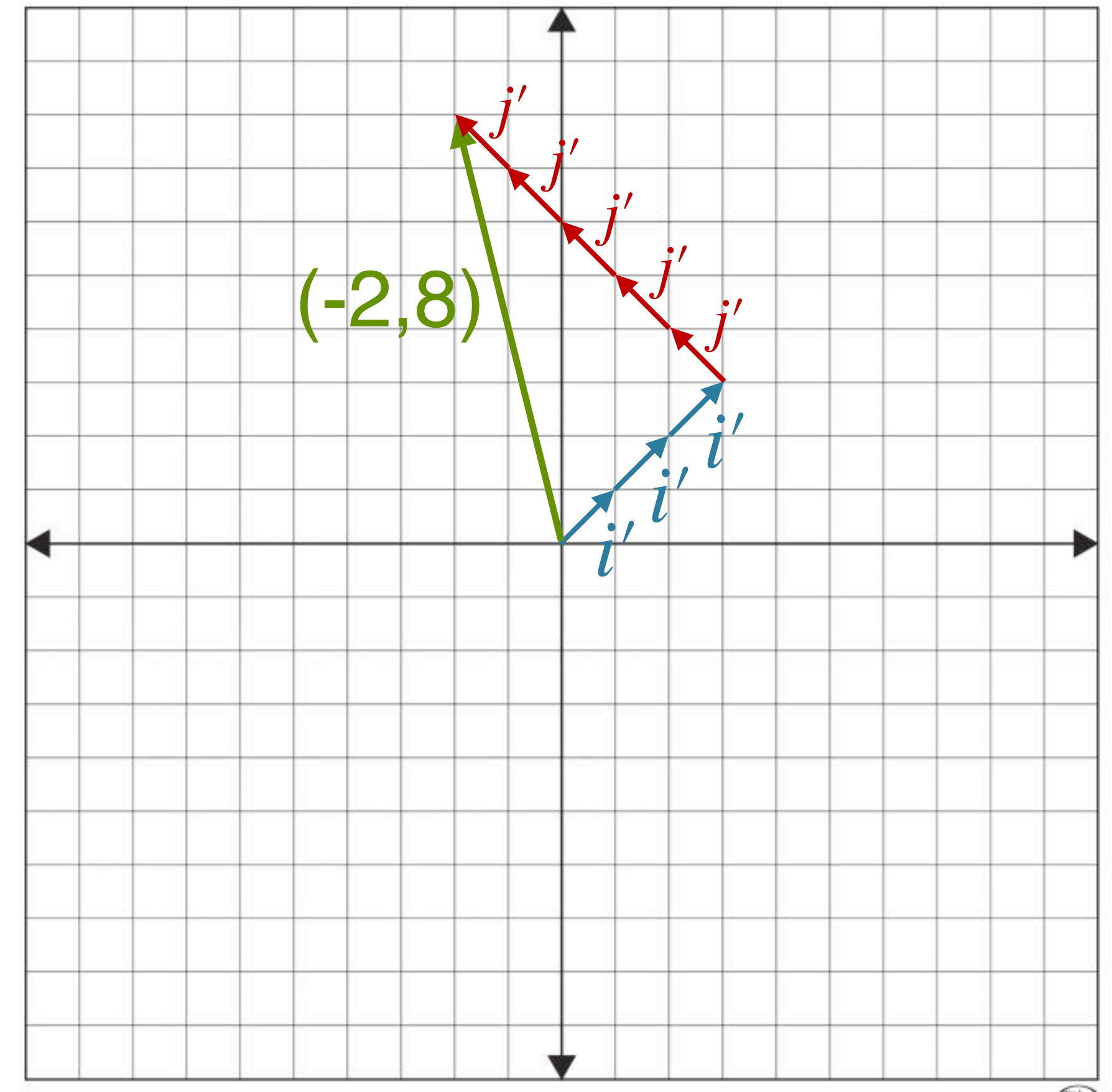


Linear Transformation

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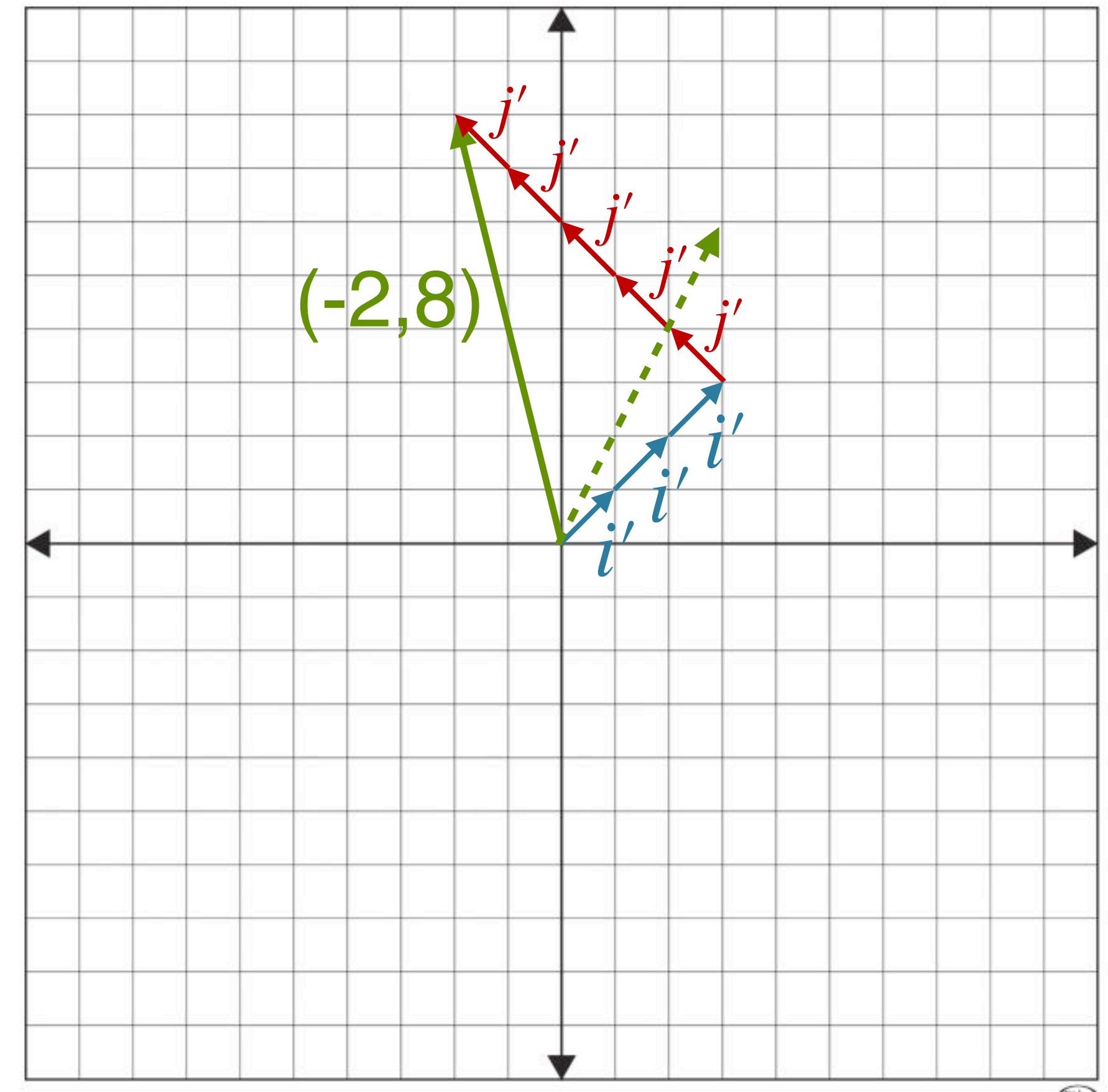


Linear Transformation

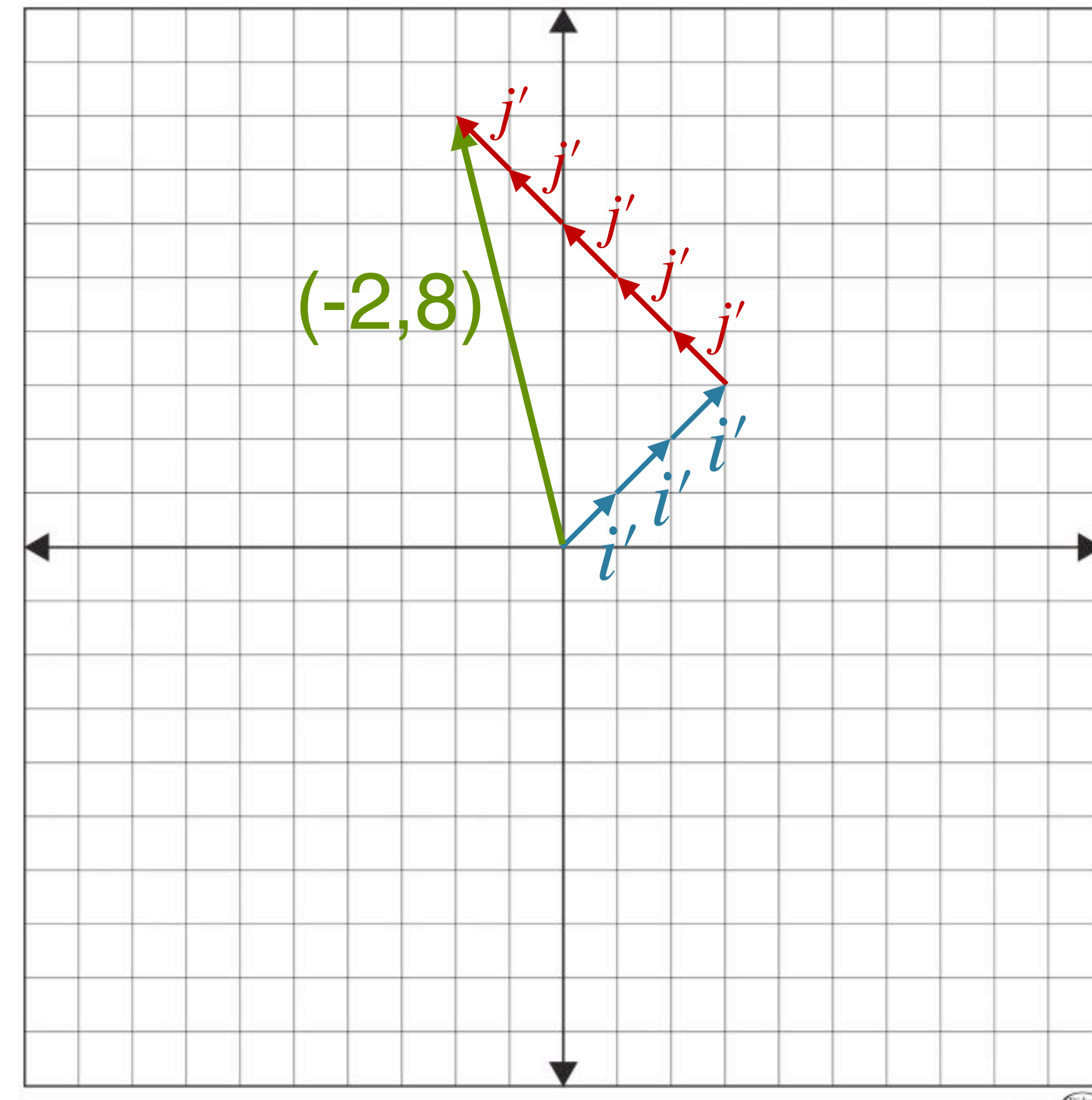
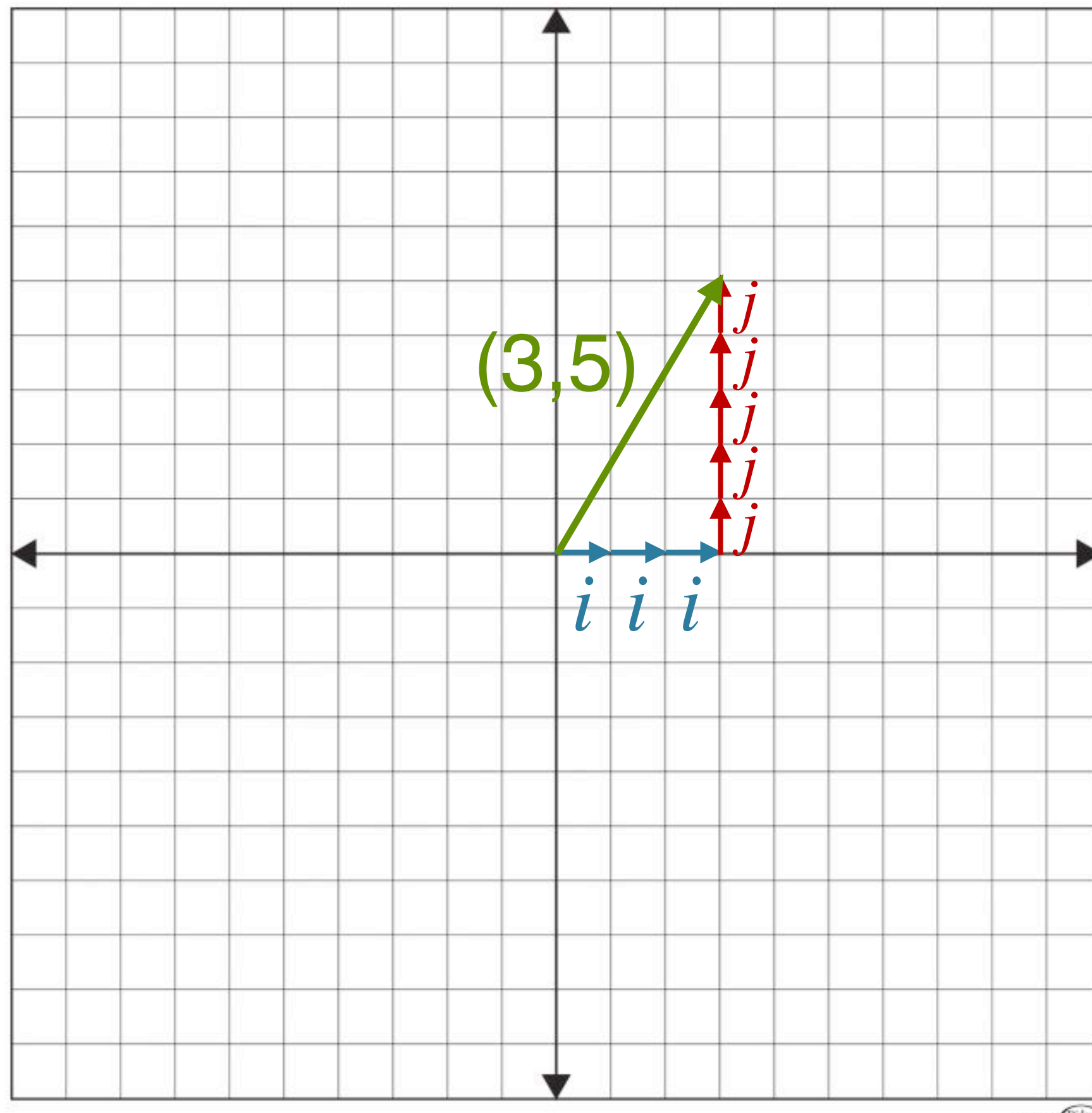
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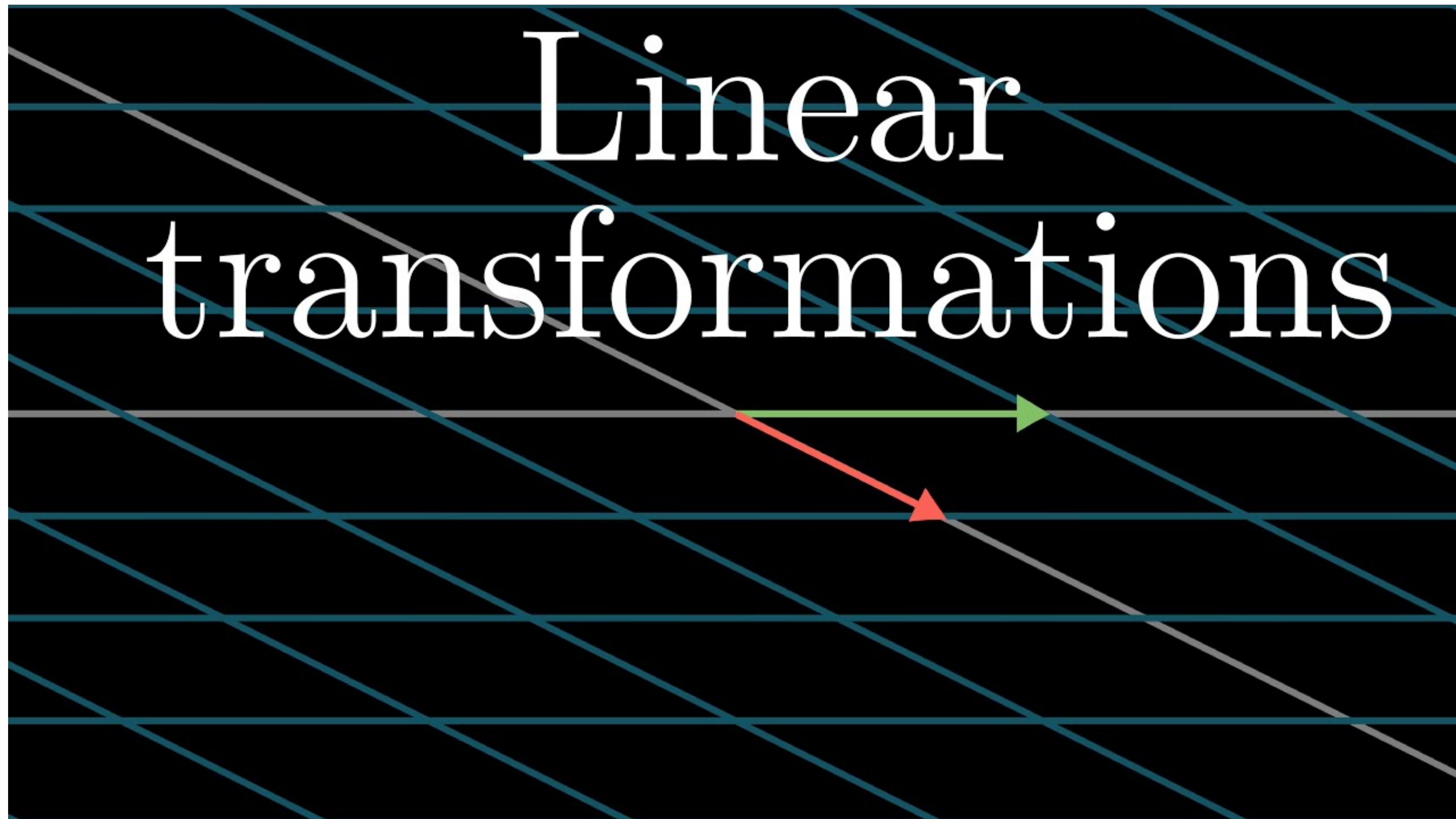
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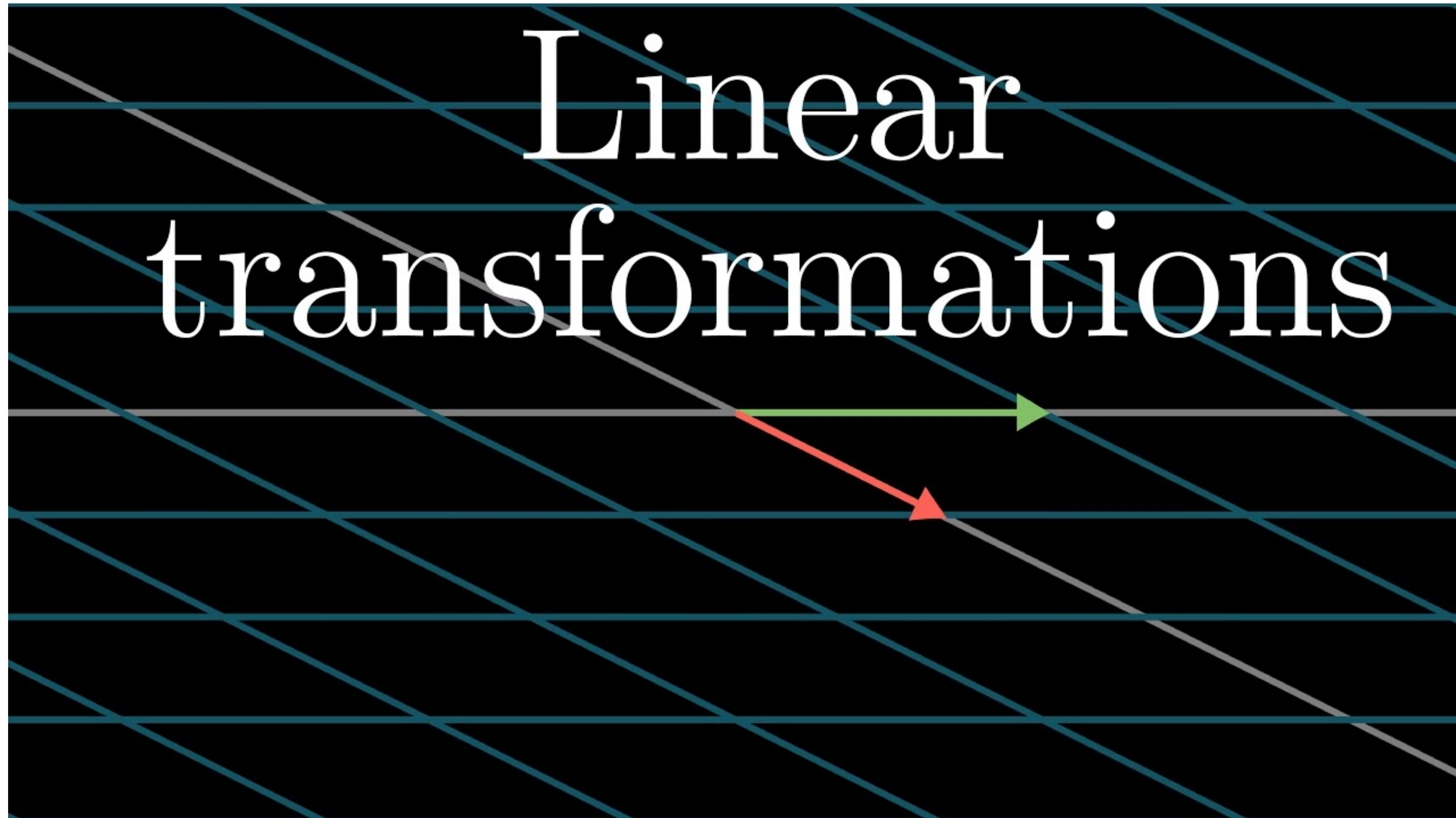
Linear Transformation



Visualizing Linear Transformations



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- TLDR: Neural Nets transform vectors and vector spaces

Quiz next session!

- You should be able to:
 - **Draw vectors** on a grid
 - **Add and subtract** vectors
 - Identify when two vectors are **independent**
 - Take the **dot product** of two vectors
 - **Multiply** a vector by a matrix
 - Give the form and discuss properties of the **Identity Matrix**
- Bring a **pen or pencil!** I will provide **calculators**